No Genuine Copula in Japanese: the pseudo-copula =*da* as an existential, conjunctive and locative complex[1]

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Abstract:

This paper proposes an alternative analysis of the Japanese so-called copula =*da* and its non-contracted form =*de aru*. While =*da* has traditionally been treated as a copula equivalent to English *be*, we argue instead that the nature of this pseudo-copula can be better understood in terms of **how Japanese, which lacks the primitive *be*, has employed its most basic linguistic resources — namely, the locative postposition =*ni*, the conjunctive suffix -*te* and the existential verb *ar*-, all of which, still today, autonomously underlie =*da* — to encode a relation between an entity and its specification as to what it is**.

Under this conception, the construction *A*=*wa B*=*de aru* — translationally 'A is B' — is analysed semantically as both **existence-introducing** and **existence-domain-specifying** (i.e. conceptual-domain-specifying). Syntactically, it consists — despite its apparent counter--intuitiveness — of the one-argument obligatory construction [*A*=*wa aru*] 'A exists' and the adjunct [*B*=*de*] ' with (A's existence) located in the conceptual domain of B' or, more practically, 'and it is as B'. Schematically, it is representable as the conjunctive proposition EXIST(A) ∧ IN(evEXIST(A), B) — whose default negation pattern is, as expected, scope-limited, namely EXIST(A) ∧ ¬IN(evEXIST(A), B), realised as *A*=*wa B*=*de*=*wa nai*, crucially with the scope-limiting =*wa* occurring after *B*=*de*.

This reanalysis makes possible a unified account of the structure, semantics and diachronic development of Japanese so-called copular constructions.

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1. Introduction — morphosyntactic and prosodic separability of *de* and *aru* in *dearu* and syntactic and prosodic constituency of *de* and the preceding NPs

In Japanese linguistics, *da/dearu* has traditionally been treated as a copula equivalent to English *be*, as in (1), where *dearu* can be replaced by *=da*, '=' indicating enclisis:

(1)	kare=wa	gakusei	dearu
	he(SUBJ)=TOP	student(PRED)	COP
	'He is a student.'		

However, this analysis faces a major challenge: it cannot account for the **high degree of independence between *de* and *aru* in *dearu***. For instance, when the noun phrase (NP) *gakusei* 'student' is marked by the focus/additive enclitic particle *=mo* 'also' — one of the so-called *toritatesi* in Japanese linguistics — the sentence cannot take the expected form shown in (2):

(2)	*kare=wa	gakusei	=mo	**dearu**
	he(SUBJ)=TOP	student(PRED)	also(ADD)	COP
	'He is also a student.'			

but must be as in (3):

(3)	kare=wa	gakusei	**de**	=mo	**aru**
	he(SUBJ)=TOP	student(PRED)	COP?	also(ADD)	COP?
	'He is also a student.'				

The particle *=mo* is inserted between *de* and *aru*, as if it were behaving as an infix from the perspective that treats *dearu* as a single copula — a situation comparable to the adverb *also* being inserted into the copula *is* in English, as in **he i-also-s a student*. Furthermore, in the complex sentence in (4):

(4)	kare=wa	gakusei	**deari**	kanozyo=wa	syakaizin	**=da**
	he(SUBJ)=TOP	student(PRED)	COP	she(SUBJ)=TOP	working adult(PRED)	COP
	'He is a student, she is a working adult.'					

the first *deari* is very frequently reduced to *de* alone, without *ari*, which is, formally, the conjunctive form — namely, the so-called *ren'yōkei* — of the existential verb *aru*, as in (5):

(5)	kare=wa	gakusei	**de**	kanozyo=wa	syakaizin	**=da**
	he(SUBJ)=TOP	student(PRED)	COP?	she(SUBJ)=TOP	working adult(PRED)	COP
	'He is a student, she is a working adult.'					

Prosodically, too, the so-called copula *dearu* exhibits unexpected behaviour. If *dearu* were a single word, it would be expected to constitute either an independent prosodic word (ω) or an enclitic, so the utterance *kare=wa gakusei dearu* would be expected to be prosodised as either (6) or (7), where " | " marks a prosodic-word boundary and prosodic words in Japanese generally correspond to traditional *bunsetu*:

(6) | *kare=wa* | *gakusei***=*dearu*** | — *dearu* as an enclitic (i.e. encliticised to *gakusei*)

(7) | *kare=wa* | *gakusei* | ***dearu*** | — *dearu* as an independent ω

In actual speech, nevertheless, the prosodisation is implemented as in (8):

(8) | *kare=wa* | *gakusei=****de* | *aru*** | — *de* as an enclitic and *aru* as an independent ω

Since prosodic-word boundaries are determined with reference to morphosyntactic information (Nespor & Vogel 2007), the fact that the utterance is prosodised as | *kare=wa* | *gakusei=de* | *aru* | strongly suggests the presence of a morphosyntactic boundary between *de* and *aru* — namely, that *de* and *aru* constitute distinct morphosyntactic elements. This prosodic separation is further supported by the fact that the discourse enclitic particle =*ne* 'you know' — one of the so-called *syūzyosi* in Japanese linguistics — can intervene between *de* and *aru*, as in (9), which would be impossible if *dearu* constituted a single prosodic word:

(9) | *kare=wa=ne* | *gakusei=****de****=ne* | ***aru****=ne* |

All of these facts demonstrate that ***de* and *aru* are morphosyntactically and prosodically separable**, whereas **the NP *gakusei* and *de* are always syntactically and prosodically inseparable**, as illustrated in (10):

(10) *Kare=wa* [*gakusei de*] [*aru*].
Kare=wa [*gakusei de*] *mo* [*aru*].
Kare=wa [*gakusei de*] [*ari*]*, kanozyo=wa syakaizin=da.*
Kare=wa [*gakusei de*]*, kanozyo=wa syakaizin=da.*
Kare=wa=ne [*gakusei de*]*=ne* [*aru*]*=ne.*

Hence, in *kare=wa gakusei dearu*, the sequence *gakusei=de* is assumed to constitute a both syntactically and prosodically defined constituent, as Dalla Chiesa (2015) notes: "the NP-*de* element must always be the phrase (in the closest proximity to the verb)"[2], where the parentheses are added by the present author. This *gakusei=de* is formally identical to that which has been treated as a postpositional phrase (PP) headed by the so-called postposition *=de*. In what follows, therefore, *gakusei=de* will be treated provisionally as a PP. This separation between *=de* and *aru* is compatible with Matsuoka's observation that "the copula *da* consists of the postposition *de* and the verb *aru*" (2019: 142). The fact that *gakusei=de* and *aru* each constitute a separate prosodic word can thus be naturally accounted for.

2. NP + *=de* as a conceptual-locative adjunct ConjP and *aru* as a one-argument existential verb

In Japanese, **phrases of the form [NP + *=de*] generally function as adjuncts**, as Hasegawa (1999: 26, translated from Japanese) states: "The case particle *=de* occurs not only for 'location of an action' but also for a variety of other elements such as 'cause' (*kaze=de yasumu* 'take time off for a cold'), 'means' (*genkin=de kau* 'buy with cash'), 'time' (*mikka=de tukuru* 'make in three days') and 'state' (*hadaka=de odoru* 'dance naked'), and all of these fundamentally function only as adjuncts. There are no predicates that require these as arguments indispensable for their informational completeness. From this, it can be generalised that *=de* marks adjuncts." Based on this generalisation, *kare=wa gakusei=de aru* can be syntactically analysed as consisting of the **obligatory argumental structure [*kare=wa aru*]** and the **adjunctive structure [*gakusei=de*]**. Further justification for analysing *gakusei=de* as an adjunct rather than an argument — despite its apparent counterintuitiveness — will be provided later in this section and in Sections 5 and 8.

In *kare=wa aru*, the verb *aru* — or, more precisely, its stem *ar-* — can thus be analysed as a **ONE-ARGUMENT predicate**, selecting only the NP *kare* as its argument. When ***ar-* functions as a one-argument predicate**, it means '**exist**', as in *sono kanōsei=wa* ***aru*** 'that possibility exists', *mada kibō=wa* ***aru*** 'the hope still exists', *sonna koto=wa* ***ari****-e-nai* 'such a thing cannot exist'

[2] She argues that the ungrammaticality of **neko=****de*** *wagahai=ga* ***aru*** 'I am a cat' supports her claim that "the NP-*de* element must always be the phrase in the closest proximity to the verb". However, the ungrammaticality of this sentence does not arise from word order, but rather from the fact that *wagahai*, while functioning as a topic, is not marked by the particle *=wa*. Indeed, although fairly awkward, *neko=****de*** *wagahai=****wa*** ***aru*** is not completely ungrammatical, and expressions such as NP=***de*** *watasi=****wa*** ***ari****-tai* 'as the referent denoted by the NP I want to exist' — e.g. *ryōsikitekina syakaizin=****de*** *watasi=****wa*** ***ari****-tai* 'as a rational member of society I want to exist' — sound much more natural.

and *sore=wa* ***at**te=wa nara-nai* 'that must not exist'. Accordingly, the sentence *kare=wa aru* can be interpreted as 'he exists', introducing *kare* as an entity. An *entity* here is understood as a *constituent of a world, actual or not*, rather than as a spatiotemporally instantiated object, and *exist* here is intended in the sense of *being an entity*. (Note that, in Contemporary Japanese, the use of *aru* as a one-argument existential verb often has a somewhat literary or philosophical flavour, as in *ware omou, yueni ware* ***ari*** 'I think, therefore I am' or ***aru****=to iu koto=wa dō iu koto=ka=to iu toi* 'the problem of what it means to exist'. This verb is more commonly used as a two-argument existential or locative verb as in *tukue=no ue=ni hon=ga* ***aru*** 'there is a book on the desk' or *hon=wa tukue=no ue=ni* ***aru*** 'the book is on the desk'. In everyday registers, the Sino-Japanese — or, *kango* — verb *sonzai-suru* is often preferred for expressing one-argument existential predication.)

This existential meaning is saliently reflected in the commonly used compound nouns ***ari****-kata* and ***ari****-yō*, which signify 'mode/manner of existing, ideal way to exist' and 'state/appearance of existing', respectively. Both are formed by combining the stem *ari-* of the verb *aru* in this sense — namely, 'exist' — with the nominal formatives *-kata* 'mode, manner, way' and *-yō* 'state, appearance', respectively. Further justification for analysing *ar-* in this construction as 'exist' — **a meaning that, for informational and pragmatic reasons, is rarely brought to the speaker's or hearer's awareness** in ordinary assertions of the meta-inferential type 'A = B' or 'A ∈ B' — will be presented in Section 6.

As for *gakusei=de*, on the other hand, although **phrases headed by =*de*** can mark, as mentioned above with reference to Hasegawa (1999: 26), a range of θ-roles such as location, cause, means, time or state, the interpretation adopted here for the present construction is **locative**, namely the primitive sense of *=de*. This use does not mark, however, a physical location but a **conceptual location**. Here, conceptual location refers to locating or anchoring an entity in the sense defined above within a conceptual space structured by domains mutually defined through semiological opposition. Consequently, *gakusei=de* means 'located or anchored in the **conceptual domain** of "student"' or, more practically, 'as a student'. A conceptual domain is a **category construed as a locus in which entities are located**. A category here is a **conceptual grouping of entities viewed as sharing common features, as constructed by human cognitive processes**. Put simply, a category here exists only as a construct of the human mind. Whether a category contains a single member or multiple ones is irrelevant. Even when *B* is a proper noun in the construction *A=wa B=de aru*, *B* signifies a conceptual domain corresponding to a category defined by the feature of being referable by that proper noun. Likewise, *kimi* 'you' in *watasi=ni totte itiban taisetuna=no=wa kimi=da* 'the most important thing for me is you' is understood as a conceptual domain defined by the category 'the unique addressee in the context of utterance or the speech situation'.

Conceptual space is founded on an **analogy with physical space**. The construction in (11):

(11) *Kare=wa syakaizin**=de** aru: sigatu=ni gakusei**=kara** syakaizin**=ni** natta.*
'He is a working adult: in April, he moved from being a student to being a working adult.'
'(lit.) He exists, **and** it is **in** the conceptual domain of "working adult": in April, he moved **from** the conceptual domain of "student" **to** that of "working adult".'

indicates that the same entity denoted by *kare* has moved within the conceptual space **from** the domain *gakusei* 'student' **to** the distinct domain *syakaizin* 'working adult', thereby relocating its existence **in** the latter domain. The postpositions *=de* 'in' in *gakusei=de*, *=kara* 'from' in *gakusei=kara* and *=ni* 'to' in *syakaizin=ni* are plausibly understood as **metaphorical extensions of the physical location, source and goal markers**, as in the construction in (12):

(12) *Kare=wa Tōkyō**=de** hataraite=iru: sigatu=ni Kyōto**=kara** Tōkyō**=ni** yatte=kita.*
'He is working in Tokyo: in April, he came to Tokyo from Kyoto.'
'(lit.) He is working, **and** it is **in** the physical domain of "Tokyo": in April, he came **from** the physical domain of "Kyoto" **to** that of "Tokyo".'

Given that **constructions encoding conceptual space systematically mirror the constructions encoding physical space** — that is, the **linguistic isomorphism between physical space and conceptual space** — it can be argued that speakers do not intentionally choose a spatial metaphor. Rather, Japanese grammar provides no alternative formal resources, and it therefore appears to formally force conceptual category domains to be treated in spatial terms.

Syntactically and semantically, the core phrasal structure found in *kare=wa gakusei=de aru* can be analysed provisionally as [$_{\mathrm{VP}_1}$ [$_{\mathrm{PP}}$ *gakusei=de*] [$_{\mathrm{VP}_0}$ *kare ar-*]]. In the lower VP [$_{\mathrm{VP}_0}$ *kare ar-*] 'he exists', the stem *ar-* of the existential verb *aru* and the NP *kare* constitute a one-argument predicate and its theme argument, respectively. In the higher VP [$_{\mathrm{VP}_1}$ [$_{\mathrm{PP}}$ *gakusei=de*] [$_{\mathrm{VP}_0}$ *kare ar-*]] 'he exists, with his existence located in the conceptual domain of "student"', on the other hand, the postposition *=de* — or, more precisely, **the underlying postposition *=ni* contained within this surface postposition–conjunction complex *=de***, as will be discussed below —

behaves as if it were a two-argument predicate, locating the eventuality 'he exists' signified by the lower VP [$_{\mathrm{VP}_0}$ *kare ar-*], within the conceptual domain signified by the adjunctive NP [*gakusei*] 'student'. Semantically, therefore, the construction expresses 'He exists, with his existence located in the conceptual domain of "student"' or, more practically, 'He exists, and it is as a student'. Since the adjunct *gakusei=de* — containing the predicate-like *=ni* within *=de* — is introduced precisely for the latter predication, **the most crucial logical** — albeit not syntactic — **predicate in this construction is** not the verbal stem *ar-* — wihch only introduces *kare*'s existence in the lower VP [$_{\mathrm{VP}_0}$ *kare* ***ar-***] — but rather **the underlying predicate-like postposition *=ni* contained within the surface postposition–conjunction amalgam *=de*** — which, in the higher VP [$_{\mathrm{VP}_1}$ [$_{\mathrm{PP}}$ $\boxed{\textit{gakusei}}$***=de***] [$_{\mathrm{VP}_0}$ $\boxed{\textit{kare ar-}}$]], locates *kare*'s existence signified by the lower VP [$_{\mathrm{VP}_0}$ *kare ar-*] in the conceptual domain signified by the NP [*gakusei*].

Schematically, using the more abstract construction ***A=wa B=de aru***, it can be represented as the conjunctive proposition **E(A) ∧ IN(evE(A), B)**. Here, the conjunct E(A) 'A exists', where E abbreviates EXIST, corresponds to the core construction *A=wa aru*, and the whole proposition E(A) ∧ IN(evE(A), B) 'A exists, with A's existence located in the conceptual domain of B' corresponds to the whole construction *A=wa B=de aru*. The core construction *A=wa aru* 'A exists', even without *B=de* 'with (A's existence) located in the conceptual domain of B', constitutes the atomic proposition E(A), since *B=de* is assumed here to be adjunctive. Thus, the first conjunct E(A) **introduces A's existence**, whereas the second conjunct IN(evE(A), B) **specifies as B the conceptual domain in which A's existence is located**. Accordingly, *A=wa B=de aru*, representable as E(A) ∧ IN(evE(A), B), means '**A exists, with A's existence located in the conceptual domain of B**'. Alternatively, using the aforementioned compound nouns, one may say that **A exists** and that **what A's existence is located in**, namely **B**, is **A's *ari-yō*** or **A's *ari-kata*** — that is, *A=ga ari, sono ari-yō/ari-kata=koso=ga B*. Thus, any essive reading of *B=de* derives from the very nature of the domain of B, rather than from *=de* itself or from *ar-*.

The notation E(A) designates schematically a linguistic unit *A ar-* constituted by its two inseparable facets — namely, the meaning 'exist- A', that is, the eventuality of A's existence, and the phonological form /ē ar-/ corresponding to it — whereas evE(A) designates only the former facet, that is, the meaning 'exist- A' or the eventuality of A's existence contained in *A ar-*: since E(A) is a predicate-functional representation of the two-facet linguistic unit *A ar-*, the operator *ev* in evE(A) — abbreviated from 'eventuality' — serves to extract only its semantic facet from E(A). Thus, IN(evE(A), B) represents **a semantic relation, *at the propositional level*, between**

an eventuality and its domain *as constructed in the proposition*, with a corresponding phonological realisation. Standard predicate-logic notation is not well suited to representing the **routine linguistic operation of locating an eventuality in a domain within the proposition** and the notation evE(A) is adopted only to make that operation explicit. Note that, since $\mathrm{IN}(\mathrm{evE}(A), B) \Rightarrow \mathrm{E}(A)$, it follows that $\mathrm{E}(A) \wedge \mathrm{IN}(\mathrm{evE}(A), B) \equiv \mathrm{IN}(\mathrm{evE}(A), B)$; hence **$\mathrm{E}(A) \wedge \mathrm{IN}(\mathrm{evE}(A), B)$ is logically redundant**, although it is still assumed to be **indispensable for a faithful formalisation of the syntactic–semantic structure of *A=wa B=de aru***.

Let us return to the adjunctivity of *B=de* in *A=wa B=de aru*. As will be discussed in Section 7, the contemporary *=de* originated historically from the sequence *=ni-te*, where *=ni* is assumed to have functioned as a locative postposition and *-te* as a conjunctive suffix. The development was probably driven by the deletion of *-i-* and by postnasal voicing — typical of the Yamato lexical stratum — yielding the sequence *=ni-te* → **=n-te* → *=n-de* → *=de*. Even today, in phrases of the form [NP + *=de*], one can observe stylistic variation between the so-called postposition *=de* and the postposition–conjunction sequence *=ni-te*, as in *Tōkyō=**de*** / *Tōkyō=**ni-te*** 'in Tokyo', though the latter is a more formal and literal form. Given this diachronic development and the synchronic variation, it is plausible to assume that the underlying form of *B=de* is still *B=ni-te*, where, strictly speaking, **it is *B=ni* headed by the underlying postposition *=ni* that marks conceptual location and the conjunctive suffix *-te* adjoins *B=ni*, not as an argument but as an adjunct**, to *A=wa aru* — or more precisely, **to the VP node [*A ar-*] within *A=wa aru***.

An NP marked by *=ni* functioning as an argument is selected by the predicate as an element semantically indispensable for completing its meaning. The link between the predicate and such an argumental *NP=ni* is determined by the argument structure inherent in the predicate, and is therefore intrinsic and necessary to it. By contrast, an underlying *NP=ni* functioning as an adjunct is optional for the predicate's meaning, and the link between the adjunct *NP=ni* and the predicate is extrinsic to and contingent on the predicate. In Contemporary Japanese, this extrinsic and contingent relation needs to be made explicit in a linguistic form, and the device used for this purpose is the conjunctive suffix *-te*. It may be assumed, from a cognitive perspective, that the *conjunctivity* of *-te* has been metaphorically extended — via the schema of *junctivity* — into *adjunctivity*, as schematised in (13):

junctivity

(13) metaphor: **conjunctivity** ------> **adjunctivity**

Having thus developed into an adjunctive suffix, *-te* adjoins *NP=ni* to the predicate or to some node in the predicate phrase as an adjunct rather than as an argument. The surface realisation of this underlying *NP=**ni-te*** is *NP=**de***. Thus, the use of *=de* as an adjunct marker in Contemporary Japanese can be accounted for by the fact that its underlying form *=ni-te* contains the conjunctive suffix *-te*, which functions adjunctively in this context. Consequently, *B=ni-te*, surfacing as *B=de*, can be analysed not as a PP but as a conjunction phrase (ConjP) headed by the conjunction *-te* with the complement PP *B=ni*, as illustrated in (14):

(14) $[_{\text{ConjP}}\ [_{\text{PP}}\ [_{\text{NP}}\ B]\ [_{\text{P}}\ {=}ni]]\ [_{\text{Conj}}\ \text{-}te]]$

In the construction *A=wa B=**de** aru*, therefore, it seems more accurate to characterise *=de* as a **complex consisting of the predicate-like conceptual-locative postposition *=ni* and the adjunctive-conjunction suffix *-te***.

The remaining problem is to determine what element in the linguistic form *A=wa B=de aru* formally corresponds to the operator ∧, which expresses logical conjunction in the proposition E(A) ∧ IN(evE(A), B). The phrasal structure corresponding to the schematic representation E(A) ∧ IN(evE(A), B) is assumed to be as in (15), in line with the analysis in (14):

(15) $[_{\text{VP}_1}\ [_{\text{ConjP}}\ B{=}ni\text{-}te]\ [_{\text{VP}_0}\ A\ ar\text{-}]]$

where the lower VP $[_{\text{VP}_0}\ A\ ar\text{-}]$ corresponds to the proposition E(A), and the higher VP $[_{\text{VP}_1}\ [_{\text{ConjP}}\ B{=}ni\text{-}te]\ [_{\text{VP}_0}\ A\ ar\text{-}]]$ — formed by adjoining the adjunct PP $[_{\text{PP}}\ B{=}ni]$ to the lower VP $[_{\text{VP}_0}\ A\ ar\text{-}]$ via the adjunctive-conjunction *-te* — corresponds to the conjunctive proposition E(A) ∧ IN (evE(A), B). The semantic composition corresponding to the structure in (15) is given in (16):

(16) $[_{\text{NP}}\ A]$:	A
$[_{\text{V}}\ ar\text{-}]$:	E( )
$[_{\text{VP}_0}\ A\ ar\text{-}]$:	E(A)
$[_{\text{NP}}\ B]$:	B
$[_{\text{P}}\ {=}ni]$:	IN(ev( ),)
$[_{\text{Conj}}\ \text{-}te]$:	∧
$[_{\text{PP}}\ B{=}ni]$:	IN(ev( ), B)
$[_{\text{ConjP}}\ B{=}ni\text{-}te]$:	∧ IN(ev( ), B)
$[_{\text{VP}_1}\ [_{\text{ConjP}}\ B{=}ni\text{-}te]\ [_{\text{VP}_0}\ A\ ar\text{-}]]$:	E(A) ∧ IN(evE(A), B)

In (16), what corresponds to the logical conjunction operator ∧, which combines the two conjuncts E(A) and IN(evE(A), B), is the adjunctive conjunction [$_{\text{Conj}}$ *-te*]. This conjunction [$_{\text{Conj}}$ *-te*] both syntactically adjoins, as stated above, the PP [$_{\text{PP}}$ *B=ni*] to the VP [$_{\text{VP}_0}$ *A ar-*] as an adjunct and, logically, functions as the conjunctive operator ∧, introducing the most crucial logical — though not syntactic — predicate in this construction IN(ev( ),), which is marked by the predicate-like postposition *=ni*. Compositionally, the lower VP [$_{\text{VP}_0}$ *A ar-*] already constitutes an independent proposition E(A) and, by merging this lower VP with the adjunctive ConjP [$_{\text{ConjP}}$ *B=ni-te*], which constitutes ∧ IN(ev( ), B), the higher VP [$_{\text{VP}_1}$ [$_{\text{ConjP}}$ *B=ni-te*] [$_{\text{VP}_0}$ *A ar-*]] is formed, constituting the conjunctive proposition E(A) ∧ IN(evE(A), B).

The syntactic derivation corresponding to (16) can be represented as in (17):

(17) *A=wa B=de aru.* '(lit.) A exists, with A's existence located in the conceptual domain B.'

Adjunct	**Core construction**
[***=ni***]	[***ar-***]
→ [***B***=*ni*]	→ [***A*** *ar-*]
→ [[*B*=*ni*] ***-te***]	
	→ **[[*B*=*ni*] *-te*]** [*A ar-*]] (adjunct embedding)
	→ [[[[*B*=*ni*] *-te*] [*A ar-*]] ***-ru***]
	→ [[[[[*B*=*ni*] *-te*] [*A ar-*]] *-ru*] ***Top***^{0}**Ø**]
	→ [***A***$_i$***=wa*** [[[[[*B*=*ni*] *-te*] [***t***$_i$ *ar-*]] *-ru*] *Top*^{0}Ø]]
	→ *A=wa B=**de** ar-u*

In (17), *-ru* is an inflexional tense suffix specified as [−past]; the category Topic, abbreviated as Top in (17), is "a kind of functional category" that "can be considered to embed the IP into discourse and context" (Hasegawa 1999: 91, translated from Japanese), and is assumed in this paper to be realised by the phonologically null morpheme *-0*, which we designate as Topic^{0}Ø, abbreviated as Top^{0}Ø. The element that moves to the specifier of TopP is assigned the particle *=wa* and functions to indicate the topic of the sentence. The underlying posposition–conjunction and verb–tense sequences *=ni-te* and *ar-ru* surface as *=de* and *ar-u*, respectively.

In opposition to this *A=wa* ***B=de*** *aru*, the semantically related and syntactically similar but distinct construction *A=wa* ***B=ni*** *aru/iru* 'A is located in B' — in which *A=wa aru/iru* 'A is located', without *B=ni* 'in B', cannot constitute a proposition, since *B=ni* is argumental, as discussed in Section 4 — can be represented as **IN(A, B)**, which represents solely the **specification of the physical domain of location**, without explicitly introducing A's existence, which is presupposed in this construction. IN(A, B) thus constitutes an **atomic proposition**. The compositional difference between the conjunctive E(A) ∧ IN(evE(A), B) and the atomic IN(A, B) is reflected in their default negation patterns, such as the partial or scope-limited negation *A=wa <u>B=de=**wa**</u> **nai*** in the former vs. the total or wide-scoped negation *A=wa <u>B=ni</u> **nai**/i-**nai*** in the latter — namely, E(A) ∧ ¬IN(evE(A), B) vs. ¬IN(A, B) — which will be discussed in Section 8.

Thus, instead of the traditional **copula-based** analysis in (18) = (1), representable as BE(kare, gakusei):

(18)	kare=wa	gakusei	**dear**-u
	he(SUBJ)=TOP	student(PRED)	**COP**-PRES
	'He is a student.'		

in this paper, a **no-copula-based** analysis in (19), representable as E(kare) ∧ IN(evE(kare), gakusei), is adopted, where CONCEP-LOC and CONJ abbreviate 'conceptual-locative' and 'conjunctive', respectively:

(19)	kare=wa	gakusei=**de** (underlyingly gakusei=**ni-te**)	**ar**-u
	he(SUBJ)=TOP	student=**CONCEP-LOC–CONJ (ADJUNCT)**	**exist**-PRES
	'He exists, with his existence located in the conceptual domain of "student".'		

Thus, the three fundamental linguistic resources employed in Japanese to encode a relation between an entity and its specification as to what it is are:

- **Predicate-like conceptual-locative postposition =*ni* 'in'**
- **Adjunctive-conjunction suffix *-te* 'and'**
- **One-argument existential verb *ar-* 'exist'**

and the conceptual device used in this paper is their compositional meaning represented by:

- **E(A) ∧ IN(evE(A), B)**

3. The so-called postposition =*de* as an Adjunctive Eventuality-Domain Specifier: ∧ IN(ev( ), NP)

In the previous section, the constituent *gakusei=de* in the construction *kare=wa* ***gakusei=de*** *aru* — representable as EXIST(kare) ∧ IN(evEXIST(kare), gakusei) — was analysed as the underlying ConjP ***gakusei=ni-te*,** which corresponds to the logical component **∧ IN(ev( ), gakusei)**. In this construction, the underlying postposition *=ni* functions conceptual-locatively, and the NP *gakusei* signifies the conceptual domain in which *kare*'s existence is located, as shown in (20):

(20) kare=wa gakusei=**de** (underlyingly gakusei=**ni-te**) ar-u
he(SUBJ)=TOP student=**CONCEP-LOC–CONJ (ADJUNCT)** exist-PRES
'He exists, with his existence located in the conceptual domain of "student".'

However, the adjunctive ConjP [NP + *=de*] can function not only conceptual-locatively, as in (20), but also **physical-locatively**, as shown in (21):

(21) kare=wa Tōkyō=**de** (underlyingly Tōkyō=**ni-te**) hataraite=i-ru
he(SUBJ)=TOP Tokyo=**PHYSIC-LOC–CONJ (ADJUNCT)** working=be-PRES
'He is working, with his working located in the physical domain of "Tokyo".'

Thus, the construction *kare=wa* ***Tōkyō=de*** *hataraite=iru* can be schematically represented as WORK(kare) ∧ **IN(ev**WORK(kare), **Tōkyō)**, in parallel to EXIST(kare) ∧ **IN(ev**EXIST(kare), **gakusei)** from *kare=wa* ***gakusei=de*** *aru*.

In both sentences, the underlying postposition *=ni*, representable as **IN(ev( ), NP)**, within the postposition–conjunction complex *=de* locates the eventualities expressed by the obligatory constructions *kare=wa hataraite=iru* and *kare=wa aru* — represented by WORK(kare) and EXIST(kare), respectively — within the domains signified by the NPs *Tōkyō* and *gakusei*, yielding additional propositions IN(evWORK(kare), Tōkyō) and IN(evEXIST(kare), gakusei). The underlying conjunction *-te*, representable as ∧, adjoins these eventuality-domain-specifying propositions to the base propositions WORK(kare) and EXIST(kare), yielding the conjunctive propositions WORK(kare) ∧ IN(evWORK(kare), Tōkyō) and EXIST(kare) ∧ IN(evEXIST (kare), gakusei). Accordingly, it can be assumed that the **Japanese postposition–conjunction complex =*de*** — underlyingly ***=ni-te*** — does not function as a simple locative postposition but can serve as what may be called an **adjunctive eventuality-domain specifier: ∧ IN(ev( ), NP)**.

4. The differences between *sore wa gakkō=DE aru* and *sore wa gakkō=NI aru*

In contrast to the construction *sore=wa gakkō=**de** **aru*** in (22), which can be analysed as **existence-introducing and existence-domain-specifying** (i.e. conceptual-domain-specifying):

(22) sore=wa gakkō=**de** (underlyingly gakkō=**ni-te**) **ar**-u
it(SUBJ)=TOP school=**CONCEP-LOC–CONJ (ADJUNCT)** **exist**-PRES
'It exists, with its existence located in the conceptual domain of "school".'

in the constructions *sore=wa gakkō=**ni** **aru*** in (23) and *kare=wa gakkō=**ni** **i**ru* in (24), analysable as **physical-location-specifying**, where PHYSIC-LOC abbreviates 'physical-locative':

(23) sore=wa gakkō=**ni** **ar**-u
it(SUBJ)=TOP school=**PHYSIC-LOC (ARGUMENT)** **be located**-PRES
'It is located in the physical domain of "school".'

(24) kare=wa gakkō=**ni** **i**-ru
he(SUBJ)=TOP school= **PHYSIC-LOC (ARGUMENT)** **be located**-PRES
'He is located in the physical domain of "school".'

the **physical-locative argument is obligatory**, as mentioned in Section 2, **and always marked by the postposition =*ni***, rather than =*de*. Thus, the verb *aru* — more precisely, its stem *ar-* — when used as a **LOCATIVE verb**, is a **TWO-ARGUMENT predicate with the argument structure [physical domain, theme]**, the location being specified by the postpositional phrase [NP + =*ni*], as in (23) and (24). In Contemporary Japanese, when the theme is animate, the verb *iru* with the stem *i-* is used instead as in (24), with the **argument PP [NP + =*ni*] unchanged**, as in *kare=wa gakkō=**ni** **i**ru* 'he is in the school'.

By contrast, the sequence of words in (25), in which the argumental PP [NP + =*ni*] in (24) has been replaced by the ConjP [NP + =*de*]:

(25) kare=wa *gakkō=**de** **i**-ru
he(SUBJ)=TOP school=LOC–CONJ (**ADJUNCT**) ?-PRES
' ? '

is semantically uninterpretable — even for native speakers — and ungrammatical, indicating that the ConjP [NP + =*de*] is adjunctive in this construction, which therefore lacks an indispensable argument marked by =*ni* not accompanied by the conjunction -*te*.

Given this general adjunctivity of the ConjP [NP + =*de*], the construction *kare=wa gakusei=de* *aru* may reasonably be regarded syntactically — though not informationally or pragmatically, as will be discussed in Section 5 — as a **structurally self-contained, existence-introducing construction** *kare=wa aru* 'he exists', **accompanied by the conceptual-locative adjunct** *gakusei=de* — underlyingly *gakusei=ni-te* — 'with (his existence) located in the conceptual domain of "student"', **which conjunctively specifies the domain of existence**.

Thus, in *A=wa B=de aru*, the constituent *B* signifies a conceptual domain of existence, whereas in *A=wa B=ni aru/iru*, *B* signifies a physical domain of location for the theme. This difference surfaces formally when B is replaced by a pro-form. As shown in (26) and (27) below, in the former construction *B* is substituted by the pro-form ***sō***, which signifies such notions as property, attribute, characteristic or manner and is glossable as 'so', whereas in the latter construction *B* is substituted by the locative demonstrative pronoun ***soko*** 'there':

(26) *A=wa* ***B****=de aru.* → *A=wa* ***sō****=de aru.*

(27) *A=wa* ***B****=ni aru/iru* → *A=wa* ***soko****=ni aru/iru.*

Although both constructions are assumed to involve a proposition of the form IN(x, y), the formal opposition between *sō* and *soko* reflects whether the second term *y* signifies a conceptual or, alternatively, a physical domain. Note that even when the postposition–conjunction complex *=de* — namely, the eventuality-domain specifier — is used, if *B* in *B=de* signifies a physical domain, it is substituted by *soko*, as in *kare=wa* ***gakkō****=de /* ***Tōkyō****=de hataraite=iru* 'he is working in a/the school / in Tokyo' → *kare=wa* ***soko****=de hataraite=iru* 'he is working there'.

5. The syntactic optionality and pragmatic necessity of *B=de* in *A=wa B=de aru*

Although *B=de* — underlyingly *B=ni-te* — in *A=wa B=de aru* is assumed to be **syntactically an adjunct**, it **cannot be deleted**. This is because, in this construction, **the most crucial logical** — though not syntactic — **predicate is** not the verbal stem *ar-* that introduces *A*'s existence, but rather **the logically predicate-like postposition *=ni* within the postposition–conjunction complex *=de*, which locates *A*'s existence in the conceptual domain signified by *B***. It is also because the sentence, while structurally consisting of the argumental *A=wa aru* 'A exists' and the adjunctive *B=de* 'with (A's existence) located in the conceptual domain of B' or, more practically, 'and it is as B', functions as a categorical or identificational sentence in which **the**

focus falls on the adjunct *B=de*.

Therefore, the **non-deletability of *B=de*** is assumed to **arise from its informational and pragmatic indispensability** rather than from syntactic necessity. The distinction can be represented as in the following table (28):

(28)

Level	**Structure**	**Function**
Syntax	[$_{\text{VP}}$ *A ar-*] + [$_{\text{ConjP}}$ *B=ni-te*]	One-argument VP + adjunctive ConjP
Semantics	'A exists, with A's existence located in the conceptual domain of B'	Existence introduction + domain specification $E(A) \wedge IN(evE(A), B)$
Information Structure	[*A=wa aru*] + [*B=ni-te*]	Given + New (default)
Pragmatics	Focus on *B=ni-te* (default)	Pragmatic obligatoriness and non-deletability of *B=de* due to focus

The verb *ar-u* is not the only predicate that requires an adjunct for informational and pragmatic reasons. Japanese has a verb meaning 'lead one's life', namely *kuras-u*. In the natural example in (29), if the physical-locative ConjP *Tōkyō=de* '(and it is) in Tokyo' is deleted, as in (30):

(29) *Kare=wa* ***Tōkyō=de*** *kurasi-te ir-u.* 'He leads his life (and it is) in Tokyo.'

(30) ?*Kare=wa kurasi-te ir-u.* 'He leads his life.'

the resulting sentence is pragmatically infelicitous. One might therefore be tempted to treat the ConjP *Tōkyō=de* as an argument. If *Tōkyō=de* were an argument, the argument structure of the stative verb *kuras-u* would be [physical domain, agent], where the agent NP refers to an entity that participates in an activity characterisable as a life. However, in (29) the ConjP *Tōkyō=de* can be replaced either by the manner AdvP *siawase-ni* 'happily' or by the comitative PP *kazoku=to* '(lit.) with the family', yielding fully natural results, as in (31) and (32):

(31) *Kare=wa* ***siawase-ni*** *kurasi-te i-ru.* 'He leads his life happily.'

(32) *Kare=wa* ***kazoku=to*** *kurasi-te i-ru.* 'He leads his life with his family.'

If one were to assume that the physical-locative ConjP *Tōkyō=de* is an argument, then the manner AdvP *siawase-ni* and the comitative PP *kazoku=to* would have to be treated as arguments. But then, what would be the θ-role instead of physical domain in [physical domain, agent]? In reality, the verb *kuras-u* is semantically self-contained in the sense of 'lead one's life' and does not inherently select any other argument than agent for semantic completeness. By default, anyone who is alive is leading a life. Presenting only this content by means of the predicate *kuras-* is therefore nearly vacuous in terms of information and pragmatics, so the hearer naturally demands some supplementary information such as physical domain, manner, comitative participant, etc. This is why (30) is felt to be infelicitous. Thus, the ConjP *Tōkyō=de* in *Tōkyō=de kuras-u* is best treated as a physical-locative adjunct rather than an argument. The same is true of the one-argument existential verb *ar-u* 'exist'. The clause *A=wa aru* 'A exists' is, on its own, informationally and pragmatically nearly vacuous in most cases, unless used in contexts such as *kanōsei=wa aru* 'the possibility exists'. Hence, this language characteristically supplies a domain specification of A's existence, yielding *A=wa B=de aru* 'A exists, with A's existence located in the conceptual domain of B', where the conceptual domain of existence is specified by *B=de*.

By contrast, the seemingly synonymous verb *sum-u* 'have one's residence (somewhere)' behaves differently. In the natural example in (33), if the physical-locative PP *Tōkyō=ni* 'in Tokyo' is deleted, as in (34):

(33) *Kare=wa* ***Tōkyō=ni*** *sun-de ir-u.* 'He has his residence in Tokyo.'

(34) **Kare=wa sun-de ir-u.* 'He has his residence.'

the result is unacceptable. (Note that the verb *sum-u* is usually translated as 'live (somewhere)'. The translation 'have one's residence' is adopted here to mirror the argument-selecting profile of *sum-u* in English.) But, unlike *kuras-u* 'lead one's life', the missing locative PP *Tōkyō=ni* in (34) cannot be substituted by either the manner AdvP *siawase-ni* 'happily' or the comitative PP *kazoku=to* 'with his family' for semantic completeness. The resultant sentences in (35) and (36) remain equally unacceptable:

(35) **Kare=wa* ***siawase-ni*** *sun-de ir-u.* 'He has his residence happily.'

(36) **Kare=wa* ***kazoku=to*** *sun-de ir-u.* 'He has his residence with his family.'

This contrast indicates that, unlike *kuras-u* 'lead one's life', *sum-u* is semantically incomplete without a locative phrase. It thus selects as an argument a physical-locative phrase specifying the place of residence. The argument structure is [physical domain, theme], unlike *kuras-u*, whose argument structure is [agent], and the locative is constituted not by the ConjP *NP=de* — underlyingly *NP=ni-te* — but by the PP *NP=ni*.

The adjunctive status of the ConjP *Tōkyō=de* in *kare=wa Tōkyō=de kurasi-te iru* in (28), where the verb *kuras-u* does not select any argument but is nevertheless informationally and pragmatically nearly vacuous on its own, and the argumental status of the PP *Tōkyō=ni* in *kare=wa Tōkyō=ni sun-de ir-u*, where *sum-u* selects a locative phrase as an argument, are consistent with the generalisation discussed in Section 2: an argumental *NP=ni* is intrinsically selected by the predicate as semantically indispensable, whereas an adjunct *NP=ni* is extrinsic to and contingent on the predicate, hence only optionally linked either to the predicate itself or to some node within the predicate phrase; this optional linkage must generally be made overt by the adjunctive-conjunction suffix *-te* in Contemporary Japanese. It follows that there are **semantic–syntactic non-deletability** and **informational–pragmatic non-deletability**, and therefore that **simple non-deletability vs. deletability is not, by itself, a reliable diagnostic for argumenthood**.

6. Shift of focus and foregrounding of the lexical meaning 'exist' of the verb *aru*

In the construction *watasi=wa ryōsikitekina syakaizin=de aru* in (37):

(37) watasi=wa ryōsikitekina syakaizin=de **ar**-u
I(SUBJ)=TOP rational member of society=CONCEP-LOC–CONJ(ADJUNCT) **exist**-PRES
'I **exist**, with my existence located in the domain of "rational member of society".'

which can be schematically represented as **E(A) ∧ IN(evE(A), B)**, the existence-introducing component E(A) — that is, the predicative function encoded in the verbal stem *ar-* 'exist' — is backgrounded as given information, whereas the existence-domain-specifying component IN(evE(A), B) is foregrounded as the informational focus of the sentence. Furthermore, as noted in Section 2, in Contemporary Japanese the verb *aru* is more commonly used with an overt locative phrase — that is, as a two-argument existential verb, as in ***B=ni*** *A=ga aru* 'there is A in/at/on B', or locative verb, as in *A=wa* ***B=ni*** *aru* 'A is in/at/on B' — than as a one-argument existential verb, as in *A=wa aru* 'A exists', and the verb *sonzai-suru* is often preferred for this type of predication, as in *A=wa sonzai-suru* 'A exists'. As a result, in ordinary assertions of the meta-inferential type 'A = B' or 'A ∈ B', the lexical meaning 'exist' is unlikely to be brought to the speaker's or hearer's awareness. By contrast, in example (38), which is representable as **A WANT [E(A) ∧ IN(evE(A), B)]** and paraphrasable as *watasi=wa ryōsikitekina syakaizin=to site sonzaisi-tai*, the situation is fairly different:

(38)	watasi=wa	ryōsikitekina syakaizin=de	**ari-**	-tai-T⁰Ø[3]
	I(SUBJ)=TOP	rational member of society=CONCEP-LOC–CONJ(ADJUNCT)	**exist**	want-PRES
	'I want to **exist**, with my existence located in the domain of "rational member of society".'			

Here, what is at issue is 'one's mode of existence', that is, an orientation toward a particular manner of being. In such a context, **E(A) is no longer backgrounded but instead becomes part of the scope of focus, as well as IN (evE(A), B)**, and **the lexical meaning 'exist' encoded in *ar-* emerges as perceptually salient**. This pragmatic shift of focus, which foregrounds the existential meaning of *ar-*, supports empirically the validity of analysing **the semantic structure of *A=wa B=de aru* not as BE(A, B) but as EXIST(A) ∧ IN(evEXIST(A), B)**.

It should be noted that the English translation *I want to BE a rational member of society*, representable as **A WANT [BECOME(A, B)]** — based on *I AM a rational member of society*, which is representable as **BE(A, B)** — is a complete mistranslation of the Japanese construction *watasi=wa ryōsikitekina syakaizin=de ARI-tai* in (38)[4]. This in itself clearly demonstrates that the Japanese *=da* or *=de aru* is not equivalent to the English *be*, nor does it function as a copula in the same way. The difference between them can be represented as in (39):

(39)	*Watasi=wa ryōsikitekina syakaizin=**de aru***:	**EXIST(A) ∧ IN(evEXIST(A), B)**
	*I **am** a rational member of society*:	**BE(A, B)**
	*Watasi=wa ryōsikitekina syakaizin=**de ari-tai***:	**A WANT [EXIST(A) ∧ IN(evEXIST(A), B)]**
	*I **want to be** a rational member of society*:	**A WANT [BECOME(A, B)]**

What corresponds in Japanese to the English *I want to be a rational member of society* in (26) is the sentence in (40):

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[3] Tense⁰Ø designates the phonologically null allomorph *-0* of the [−past] tense suffix, which is canonically realised as *-ru*, as noted in Section 10.

[4] In idiomatic English, (38) can be translated more naturally, albeit not perfectly, as 'I want to LIVE AS a rational member of society'.

(40) *Watasi=wa ryōsikitekina syakaizin=**ni nari-tai***: **A WANT [BECOME(A, B)]**

which can be represented as shown in (41), containing the two-argument state-change verb *naru* 'become', with *=ni* marking the argument GOAL:

(41)				
	watasi=wa	ryōsikitekina syakaizin=ni	**nari-**	-tai-T^{o}Ø
	I(SUBJ)=TOP	rational member of society=GOAL(ARGUMENT)	**become**	want-PRES
	'I want to BE a rational member of society.'			

Finally, building on the example in (38), one can demonstrate that the ConjP *B=de* in *A=wa* ***B=de*** *ar-u* — which constitutes the propositional core of the desiderative sentence *A=wa* ***B=de*** *ari-tai* — is not an argument but an adjunct. In the natural example in (42) = (38), if the social-role-specifying ConjP *ryōsikitekina syakaizin=de* '(and it is) as a rational member of society' is deleted, as in (43):

(42) *Watasi=wa* [*ryōsikitekina syakaizin=de*] *ari-tai.* '(lit.) I want to exist as a rational member of society.'

(43) ?*Watasi=wa ari-tai.* '(lit.) I want to exist.'

the resulting sentence is pragmatically infelicitous. One might thus be tempted to treat the ConjP *ryōsikitekina syakaizin=de* as an argument. However, in (42) this ConjP can be replaced by the character-specifying verbal-adjectival stem *kedaka-ku* 'nobly', which is used adverbially, the psychological-stative nominal-adjectival stem *kokoro-odayaka-ni* 'peaceful-mindedly' also employed adverbially, the psychological-state-specifying ConjP selecting a nominal-adjectival phrase as its complement *kokoro-odayaka-de* '(and it is) as being peaceful-minded' or the ConjP with an interpersonal-relation-specifying complement PP *tuma=to(=wa) ninin-sankyaku=de* '(and it is) hand in hand with my wife', yielding completely natural results, as in (44) – (47):

(44) *Watasi=wa* [*kedaka-ku*] *ari-tai.* '(lit.) I want to exist [nobly].'

(45) *Watasi=wa* [*kokoro-odayaka-ni*] *ari-tai.* '(lit.) I want to exist [peaceful-mindedly].'

(46) *Watasi=wa* [*kokoro-odayaka-de*] *ari-tai.* '(lit.) I want to exist [as being peaceful-minded].'

(47) *Watasi=wa* [*tuma=to(=wa) ninin-sankyaku=de*] *ari-tai.* '(lit.) I want to exist [hand in hand with my wife].'

Since the categorially heterogeneous constituents found in these brackets [ ] cannot all be assigned the same θ-role, they are all adjuncts that specify the manner of existence introduced by the one-argument existential verb *ar-*. Put differently, the sentences in (42) and (44) – (47) may be answers to the manner question *watasi/anata=wa **dō** ari-tai(=no)=ka* '**How** do I/you want to exist?' As stated in the previous chapter, their non-deletability appears to result from the informational and pragmatic vacuousness of the adjunct-deleted sentence *watasi=wa ari-tai*.

7. From Old Japanese to Contemporary Japanese forms

In Old Japanese (OJ, 700–800), the modern formally distinct constructions *A=wa B=**de aru*** 'A exists, with A's existence located in the conceptual domain of B' and *A=wa B=**ni aru/iru*** 'A is located in the physical domain of B' would have corresponded to the seemingly identical construction *A B=**ni ari***. "Dative *ni* is the general oblique case, marking both argument and non-argument oblique nominals" (Frellesvig 2010: 126), unlike in Contemporary Japanese, where *=ni* marks arguments, while *=de* — which originated from the postposition–suffix sequence *=nite* — marks adjuncts, in accordance with the diachronic description "OJ *ni* and *nite* became more specialized with *ni* being used more for arguments and *nite* more for adjuncts; from mid to late EMJ *nite* acquired the variant *de* which is still in use in contemporary NJ" (*ibid.*: 243), where EMJ and NJ abbreviate Early Middle Japanese (800–1200) and 'new Japanese' — namely, Modern Japanese, 1600– — respectively.

Thus, the former *A B=ni ari* can be analysed as an existence-introducing and existence-domain-specifying construction consisting of both the argumental structure *A ari* 'A exists' — with the verb *ari* being used as **one-argument existential predicate** — and the conceptual-locative adjunct *B=ni* '(A's existence) located in the conceptual domain of B', as shown in (48). The latter *A B=ni ari* is, on the other hand, analysable as a physical-location-specifying construction consisting solely of the argumental structure *A B=ni ari* 'A is located in the physical domain of B', in which the verb *ari* is employed as a **two-argument locative predicate** and *B=ni* constitutes the physical-locative argument, as indicated in (49):

(48)	A	B=**ni**	**ari**-T^{0}Ø
	A(SUBJ)=TOP	B=CONCEP-LOC(**ADJUNCT**)	**exist**-PRES

'A exists, A's existence located in the conceptual domain of B.'

(49) A B=**ni** **ari**-T^{0}Ø
A(SUBJ)=TOP B=PHYSIC-LOC(**ARGUMENT**) **be located**-PRES
'A is located in the physical domain of B.'

The Old Japanese constructions in (48) and (49) appear superficially identical, but when the element *B* is substituted by a pro-form, they diverge as in (50) and (51):

(50) *A* ***B****=ni ari.* → *A* ***sa****=ni ari.* (conceptual domain)

(51) *A* ***B****=ni ari.* → *A* ***soko****=ni ari.* (physical domain)

Consequently, although both constructions is assumed to involve a proposition of the form IN(x, y), the pro-form opposition between *sa* and *soko* reflects whether the second term *y* refers to a conceptual domain or to a physical domain.

These superficially identical constructions in (48) and (49), however, diverged diachronically into the formally distinct modern constructions in (52) and (53):

(52) A=wa B=**ni-te** (surfacing as B=**de**) **ar**-u
A(SUBJ)=TOP B=CONCEP-LOC–CONJ(**ADJUNCT**) **exist**-PRES
'A exists, with A's existence located in the conceptual domain of B.'

(53) A=wa B=**ni** **ar**-u / **i**-ru
A(SUBJ)=TOP B=PHYSIC-LOC(**ARGUMENT**) **be located**-PRES
'A is located in the physical domain of B.'

As already stated in Section 4, in Contemporary Japanese as well, the element *B* undergoes different pro-form substitutions, as in (54) = (26) and (55) = (27):

(54) *A=wa* ***B****=de aru.* → *A=wa* ***sō****=de aru.* (conceptual domain)

(55) *A=wa* ***B****=ni aru/iru* → *A=wa* ***soko****=ni aru/iru.* (physical domain)

In (53), the locative verbs *aru* and *iru*, both semantically corresponding to the Old Japanese *ari* — though the latter originally derived from another verb *wiru* 'sit; be seated' — are used for inanimate and animate themes, respectively. The historical development from (48) to (52) is as follows. In Old Japanese, there existed "analytic forms consisting of an infinitive (*ni* …) followed by *ar*-" (Frellesvig 2010: 95) for the so-called copula — that is, *ni ar*-. In the subsequent Early Middle Japanese (EMJ, 800–1200) period, this analytic form "*ni ar*- had the variant *nite ar-*" (*ibid.*: 235). "In the course of EMJ" (*ibid.*: 235), or more specifically "from the end of the period" (*ibid.*: 233), "*nite ar-* became *de ar-*" (*ibid.*: 235). As illustrated in "Table 12.8. *Development of the copula and adjectival-copula paradigms*" (*ibid.*: 344), from the end of EMJ to early Late Middle Japanese (early LMJ, Kamakura period 1185-1333), the three forms *ni ar-*, *nite ar-* and *de ar-* coexisted, although "already from late EMJ" (Insei period, 1086-1185), "*ni* was getting more restricted" and "*de* was used far more widely" than *nite*. The same table (*ibid.*: 344) shows that only the newest *de ar-* continued to be used from late LMJ (Muromachi period 1333-1573) into Contemporary Japanese.

The historical developments in (56) and (57):

(56) *A B=**ni** ari* → *A B=**ni-te** ari* → *A=wa B=**de** aru* — E(A) ∧ IN(evE(A), B)

(57) *A B=**ni** ari* → *A B=**ni** ari* → *A=wa B=**ni** aru/iru* — IN(A, B)

may plausibly reflect the diachronic split between *=ni*, used for arguments, and *=ni-te/=de* used for adjuncts, where the conjunctive suffix *-te* adjoins the PP [NP + *=ni*] as an adjunct, rather than as an argument, to another node, as discussed in Section 2. This suggests a shift from a stage in which *=ni* marked both arguments and adjuncts, without the help of the conjunction *-te*, to one in which *=ni* and *=ni-te/=de* became specialised as argumental and adjunctive markers, respectively. Thus, the historical split of the Old Japanese *=ni ari* between the Modern Japanese *=de aru* and *=ni aru/iru* may not reflect a reanalysis of a single construction, but rather the formal divergence of two distinct semantic and syntactic structures — 'existence and its location in a conceptual domain' vs. 'location in a physical domain' — that happened to share the same surface form *A B=ni ari* in Old Japanese — a possibility that merits further investigation.

8. Default partial negation and the scope of negation marked by the particle =*wa*

The **default negation pattern of the existence-introducing and existence-domain-specifying** sentence in (58) = (19):

(58)	kare=wa	gakusei=de	ar-u
	he(SUBJ)=TOP	student=CONCEP-LOC–CONJ(ADJUNCT)	exist-PRES
	'He exists, and it is as a student.'		

which can be schematically represented as the conjunctive proposition E(A) ∧ IN(evE(A), B):is **partial**, or **scope-limited**, as in (59):

(59)	kare=wa	gakusei=de=**wa**	(*ara-)	**-nai**-T^{0}Ø
	he(SUBJ)=TOP	student=CONCEP-LOC–CONJ(ADJUNCT)=**SCOPE OF NEG**	exist	**NEG**-PRES
	'He exists, but it is **not** as a student.'			

rather than total, or wide-scoped, in (60):

(60)	kare=wa	gakusei=de	(*ara-)	**-nai**-T^{0}Ø
	he(SUBJ)=TOP	student=CONCEP-LOC–CONJ(ADJUNCT)	exist	**NEG**-PRES
	'He does **not** exist as a student.'			

(Note that in (59) and (60), the stem of the existential verb *aru* is represented as (*ara-), since the negative form of this verb is today not the theoretically expected **ara-nai* — with the so-called *mizenkei* stem *ara-* — but obligatorily the suppletive adjective *nai*.) The more natural sentence *kare=wa gakusei=de=**wa nai*** in (59) — or, more colloquially, *kare=wa gakusei=dy**a** **nai***, with the contracted form *=dya* derived from the sequence *=de=wa* — does **not negate the existence of *kare* itself**, but rather **negates only *gakusei=de* as the relevant domain of *kare*'s existence**. In this way, **the particle =*wa* can mark the scope of negation in Japanese**.

Schematically, the partial negation *A=wa B=de=**wa nai*** in (59) — underlyingly, **A=wa B=de=**wa** ara-**nai*** — can be represented as **E(A) ∧ ¬IN(evE(A), B)**, whereas the total negation *A=wa B=de **nai*** in (60) — underlyingly, **A=wa B=de ara-**nai*** — can be represented as **¬(E(A) ∧ IN(evE(A), B)) ≡ ¬E(A) ∨ ¬IN(evE(A), B)**. Although this disjunction is compatible with ¬E(A) ∧ IN(evE(A), B), E(A) ∧ ¬IN(evE(A), B) or ¬E(A) ∧ ¬IN(evE(A), B), the first case is inconsistent with the conditional IN(evE(A), B) ⇒ E(A) — hence, the contrapositive ¬E(A) ⇒ ¬IN(evE(A), B) — and the third case, ¬E(A) ∧ ¬IN(evE(A), B) ≡ ¬E(A), is pragmatically infelicitous in topicalised constructions with *A=wa*, since *A=wa*, as a topicalised phrase, tends to presuppose the contextual or situational givenness — that is, the existence — of *A* and, moreover, since there are dedicated Japanese forms for existential negation such as *A=wa nai / i-nai / sonzai-si-nai*; in practice, therefore, the interpretation converges on E(A) ∧ ¬IN(evE(A), B). Hence, the former pattern in (59), which unambiguously expresses the partial negation E(A) ∧ ¬IN(evE(A), B), is judged to be the more natural one.

This analysis is further supported by Kansai variety forms such as *gakusei=ya nai* and *gakusei=ya ara-hen*, which are natural and frequent and correspond closely to the standard forms *gakusei=de=wa nai* and *gakusei=dya nai*. In the former forms, the focus particle *=ya* originated from the sequence *=de=wa*, via *=dya*, and *-hen* functions as a negator, just as *-nai* does. By contrast, corresponding forms without this particle — namely, *gakusei=de nai* and *gakusei=de ara-hen* — are pragmatically infelicitous.

If *=de aru* were a single copula, then *A=wa B=de aru* would represent BE(A, B), and its negation would take the form of simple predicate negation *A=wa B=de nai*, representable as ¬BE(A, B), with no *=wa* following *=de*. **Empirically, however, *A=wa B=de nai* is perceived as infelicitous, whereas *A=wa B=de=wa nai* and *A=wa B=dya nai* are preferred**. This preference for the particle *=wa* after *=de* is unexpected under the simple-copula analysis, but follows naturally if the sentence is decomposed as E(A) ∧ IN(evE(A), B), so that the logically and pragmatically most plausible interpretation is the partial negation E(A) ∧ ¬IN(evE(A), B).

By contrast, the **default negation pattern of the physical-location-specifying** sentences in (61) = (23) and (62) = (24), which can both be represented as IN(A, B):

(61)	sore=wa	gakkō=ni	ar-u
	it(SUBJ)=TOP	school=PHYSIC-LOC(ARGUMENT)	be located-PRES
	'It is located in the school.'		

(62)	kare=wa	gakkō=ni	i-ru
	he(SUBJ)=TOP	school= PHYSIC-LOC(ARGUMENT)	be located-PRES
	'He is located in the school.'		

is **total**, as indicated in (63) and (64):

(63)	sore=wa	gakkō=ni	(*ara-)	**nai**-T^{0}Ø
	it(SUBJ)=TOP	school=PHYSIC-LOC(ARGUMENT)	be located	**NEG**-PRES
	'It is **not** located in the school.'			

(64)	kare=wa	gakkō=ni	i-	**nai**-T^{0}Ø
	he(SUBJ)=TOP	school= PHYSIC-LOC(ARGUMENT)	be located	**NEG**-PRES
	'He is **not** located in the school.'			

Unlike the existence-introducing and existence-domain-specifying conjunctive proposition E(A) ∧ IN(evE (A), B), the physical-location-specifying IN(A, B) is atomic and, naturally, its negation constitutes ¬IN(A, B), which corresponds to (63) and (64).

On the other hand, when the PP *gakkō=ni* in (63) and (64) is marked by *=wa*, this particle functions contrastively, as in (65) and (66), respectively, where CONTR abbreviates contrastive:

(65)	sore=wa	gakkō=ni=**wa**	(*ara-)	**nai**-T^{0}Ø
	it(SUBJ)=TOP	school=PHYSIC-LOC(ARGUMENT)=**CONTR**	be located	**NEG**-PRES
	'It is **not** located in the school **but somewhere else**.'			

(66)	kare=wa	gakkō=ni=**wa**	i-	**nai**-T^{0}Ø
	he(SUBJ)=TOP	school= PHYSIC-LOC(ARGUMENT)=**CONTR**	be located	**NEG**-PRES
	'He is **not** located in the school **but somewhere else**.'			

In contrast to (63) and (64), which are representable as $\neg$IN(A, B), (65) and (66) can be represented schematically as $\neg \mathrm{IN}(A, B) \wedge \exists x\ (\mathrm{IN}(A, x) \wedge x \neq B)$ — semantically, 'A is not located in B, and there exists a physical domain x distinct from B such that A is located in x'.

The difference in default negation pattern between the existence-introducing and existence-domain-specifying construction and the physical-location-specifying one lies in the fact that the former, **E(A) $\wedge$ IN(evE(A), B), is conjunctive**, whereas the latter, **IN(A, B), is atomic**. This ultimately derives from **the status of B as an adjunct in the former and as an argument in the latter**, since if B were an argument, E(A) $\wedge$ IN(evE(A), B) would correspond to something like EXIST-MANNER(A, B), whose negation would be the total $\neg$EXIST-MANNER(A, B).

Finally, if *=de aru* were a single copular unit, partial negation with *=wa* would necessarily take the form **kare=wa <u>gakusei=**wa**</u> de **nai***, as in (67):

(67)	*kare=wa	gakusei=**wa**	de	**nai**-T^{0}Ø
	he(SUBJ)=TOP	student(PRED)=**CONTR**	COP	**NEG**-PRES
	'He is **not** a student **but something else**.'			

This ungrammaticality further supports the separability of *=de* and *aru* and the constituency of *gakusei=de*.

9. The differences between *A=wa B=de aru* and *A=ga B=de aru*

The following table (68) provides a preliminary overview of the propositional identity and the information-structural differences between sentences of the type ***A=wa B=de aru*** and those of the type ***A=ga B=de aru***, without attempting a full formalisation, which lies beyond the scope of this paper:

(68)

Sentence	Propositional structure	Informational structure	
		Given	New
*A=**wa** B=de aru*	E(A) ∧ IN(evE(A), B)	E(A)	IN(evE(A), B)
*A=**wa** B=de=wa nai*	E(A) ∧ ¬IN(evE(A), B)	E(A)	¬IN(evE(A), B)
*A=**wa** B=de=wa naku* *C=**wa** B=de aru*	E(A) ∧ ¬IN(evE(A), B) ∧ E(C) ∧ IN(evE(C), B)	E(A) ∧ E(C)	¬IN(evE(A), B) ∧ IN(evE(C), B)
*A=**ga** B=de aru*	E(A) ∧ IN(evE(A), B)	∃x (E(x) ∧ IN(evE(x), B))	x = A
*A=**ga** B=de=wa nai*	E(A) ∧ ¬IN(evE(A), B)	∃x (E(x) ∧ IN(evE(x), B))	x ≠ A
*A=**ga** B=de=wa naku* *C=**ga** B=de aru*	E(A) ∧ ¬IN(evE(A), B) ∧ E(C) ∧ IN(evE(C), B)	∃x (E(x) ∧ IN(evE(x), B))	(x ≠ A) ∧ (x = C)

Syntactically, the structure of *A=**ga** B=de aru* in (68) can be represented as in (69):

(69) $[_{TP}\ \boxed{[_{NP}\ A_i{=}ga]}\ [_{T'}\ \boxed{[_{VP_1}\ [_{ConjP}\ B{=}ni\text{-}te]\ [_{VP_0}\ [_{NP}\ x_i]\ [_{V}\ ar\text{-}]]]}\ [_{T}\ \text{-}ru]]]$

where *x* designates a trace, used here instead of the conventional notation *t*. In this structure, the higher VP $[_{VP_1}\ [_{ConjP}\ B{=}ni\text{-}te]\ [_{VP_0}\ [_{NP}\ x_i]\ [_{V}\ ar\text{-}]]]$ conveys the given information ∃x (E(x) ∧ IN(evE(x), B)), whereas the **untopicalised** — and thus capable of indicating a piece of new information — NP $[_{NP}\ A_i{=}ga]$ conveys the new information x = A, through the anaphoric relation of the trace $[_{NP}\ x_i]$ to its antecedent $[_{NP}\ A_i{=}ga]$. Note that the notation ∃x here indicates that there exists a variable x which can be instantiated by a constant as new information within the information structure, whereas E(x) indicates that the linguistic sign to which x is anaphoric refers to an entity, understood here as a constituent of a world, whether actual or non-actual, as noted in Section 2. The constituents conveying given and new information are enclosed within boxes and, furthermore, those encoding new information are shaded. (The bracket notation adopted here is a descriptive representation of hierarchical constituency, and the traditional category labels are used merely as a device for visualising the structure.)

On the other hand, the structure of *A=**wa** B=de aru* in (68) can be represented as in (70):

(70) $[_{TopP}\ \boxed{[A_i{=}wa]}\ [_{Top'}\ [_{TP}\ [_{VP_1}\ \boxed{[_{ConjP}\ B{=}ni\text{-}te]}\ \boxed{[_{VP_0}\ [_{NP}\ x_i]\ [_{V}\ ar\text{-}]]}]\ [_{T}\ \text{-}ru]]\ [_{Top}\ Top^0\varnothing]]]$

In this structure, the **topicalised** phrase $[A_i{=}wa]$ — which typically conveys a piece of given information, since it typically refers to an entity already introduced in the discourse, namely a referent already known to the addressee — and the lower VP $[_{VP_0}\ [_{NP}\ x_i]\ [_{V}\ ar\text{-}]]$ — where the verb $[_{V}\ ar\text{-}]$ 'exist' provides another piece of given information, since the topicalised phrase $[A_i{=}wa]$ generally presupposes the existence of *A* — together convey the given information

E(A). By contrast, the ConjP [$_{\text{ConjP}}$ *B=ni-te*] encodes the new information ∧ IN(evE(A), B), since a sentence conveying solely given information would be informationally vacuous. Note that, since IN(evE(x), B) ⇒ E(x), it follows that ∃x (E(x) ∧ IN(evE(x), B)) ≡ ∃x IN(evE(x), B); nevertheless, the logically redundant ∃x (E(x) ∧ IN(evE(x), B)) represents a component within the informational structure that corresponds to the syntactic–semantic structure assumed in this paper.

10. The contracted form *=da*

In spoken language, the contracted form *=da* is very often used instead of *=de aru*, which is more formal and literal, as in *kare=wa gakusei***=da** rather than *kare=wa gakusei***=de aru**. It should be noted that this **contracted form *=da*** does not represent a new copular category 'be' but is a **purely phonological reduction of the postpositional–conjunctive–verbal complex *=de aru***, as shown in (71):

(71)	kare=wa	gakusei=**d**	**a**-T⁰Ø-Top⁰Ø
	he(SUBJ)=TOP	student=**CONCEP-LOC–CONJ**(ADJUNCT)	**exist**-PRES
	'He exists, with his existence located in the conceptual domain of "student".'		

The historical change from *=de aru* to *=da*, possibly via *=dea*[5], involves no semantic or syntactic reanalysis, but a process of phonological reduction comparable to German *in dem* → *im* 'in the' or Portuguese *de a* → *da* 'from the' or 'of the'. Just as there is no semantic necessity for *in* and *dem* to fuse into *im*, or for *de* and *a* into *da* — a phenomenon comparable to the English *in the*, *from the* or *of the* forming a single word — there is likewise no semantic motivation for *=de* and *aru* to form a SINGLE lexical item.

That said, however, the reduction of *aru* to *a* can be accounted for in informational and pragmatic terms. First, as already dicussed in Sections 2 and 5, the crucial logical — though not syntactic — predicate in this construction is not the verbal predicate *ar-*, representable as E( ), but the underlying predicate-like postposition *=ni*, representable as IN(ev( ),), contained within the surface postposition–conjunctive complex *=de*.

[5] Historically, the form *=dea*, which yielded the modern *=da*, can be attested in Azuchi–Momoyama-period materials: *bonnin yorimo giŭzaini fuxôzuru coto***dea** "(it) is a reason for imposing a heavier guilt (on him) than on an ordinary person" (https://www2.ninjal.ac.jp/textdb_dataset/en/amis/amis-rk-001.html: Amakusa Edition of *Esopo no fabulas*, 1593, p.475, ll.10-11, in Text Data Sets for Research on the History of Japanese).

Second, the topicalised phrase *A=wa*, marked by the topicalisation particle *=wa*, generally presupposes at the discourse level the existence of *A*, namely E(A). Moreover, the adjunctive-conjunction phrase *B=de* — underlyingly *B=ni-te* — is representable as the additional conjunct ∧ IN(ev( ), B). When *A=wa* and *B=de* occur successively in the string *A=wa B=de…*, the hearer is led to anticipate the full conjunctive proposition E(A) ∧ IN(evE(A), B). In this context, the existential verb *aru* becomes informationally redundant when it once again introduces E(A).

In non-sentence-final coordinate clauses, the inflected form *ari* of *aru* in *=de ari* is often omitted, which is a natural consequence of the above anticipation. Returning to Section 1, the compound sentence *kare=wa gakusei=de, kanozyo=wa syakaizin=d**a*** 'he (is) a student, and she is a working adult' in (5) — underlyingly, *kare=wa gakusei=de* ***ari****, kanozyo=wa syakaizin=d**a*** — can be analysed as in (72):

(72)	kare=wa	gakusei=de	(**ari**)	kanozyo=wa	syakaizin=d	**a**-T^{0}Ø-Top^{0}Ø
	he(SUBJ)=TOP	student=LOC-CONJ	(**exist**)	she(SUBJ)=TOP	working adult=LOC-CONJ	**exist**-PRES

'He (**exists**), with (his **existence**) located in the conceptual domain of "student", she **exists**, with her **existence** located in the conceptual domain of "working adult".'

The construction in (72) is an instance of zeugma, in which the inflected form *ari* of the existential verb *aru* is deleted, since, as stated in the previous paragraph, the proposition E(kare) ∧ IN(evE(kare), gakusei) is predictable already at the point *kare=wa gakusei=de*, and since *ari* in *gakusei=de ari* is contextually recoverable on the basis of the structural parallelism between the two coordinate clauses. Therefore, *=de* is not an inflected form of *=da*.

Third, in a sentence in which IN(evE(A), B) is presented as new information, E(A) is presupposed and thus treated as given, owing to the conditional IN(evE(A), B) ⇒ E(A). In this context, E(A) is pragmatically and informationally less significant than the focus component IN(evE(A), B).

It may therefore be assumed that **this low informational and pragmatic salience is reflected in the phonological reduction of *aru* to *a***. This reduction can be analysed as follows: morphologically, the [−past] tense suffix has the phonologically null allomorph /-0/ — which we designate as Tense^{0}Ø, abbreviated as T^{0}Ø — instead of the canonical *-u* /-ru/ in *ar-u* /ar-ru/, yielding /ar-0/; phonologically, the word-final /-r/ is deleted in phonetic realisation, giving rise to the surface form *a*.

When the topic marker *=wa* is used, as in *A=wa* in Contemporary Japanese sentences of the type *A=wa B=de aru* or *A=wa B=da*, *A* is immediately understood, at the point of *A=wa*, to be the topic, and the existence of the referent of *A* is generally discourse-presupposed at that point. Consequently, in the non-contracted form *A=wa B=de aru*, introducing its existence once again by means of the sentence-final *aru* is informationally redundant. By contrast, in a form such as Old Japanese *A B=ni ari*, it is not formally indicated whether *A* is topicalised or not, and the introduction of the existence of the referent of *A* by the sentence-final *ari* is therefore not informationally redundant. It is thus possible that the expansion of the use of the particle *=wa* as a topic marker and the reduction *aru → a* in *=de aru* are correlated.

In this respect, Kitazaki (2024: 486; my translation from Japanese) — a study that constructs a parallel corpus of the *Kōya* manuscript of *Heike monogatari*, composed in early Late Middle Japanese (early LMJ, Kamakura period, 1185–1333), and the *Amakusa* edition of *Heike monogatari*, completed in 1592 as its colloquial translation in late Late Middle Japanese (late LMJ, Muromachi period, 1333–1573), and uses this parallel corpus to compare corresponding passages between the two texts — states: "When the two texts are compared, one frequently notices correspondences in which the particle *wa* is added in the *Amakusa* edition to zero-marked noun phrases (indicated by ∅) that function as topics in the *Kōya* manuscript. ... There are many examples in which *wa* is newly added in the *Amakusa* edition, whereas there are few examples in which *wa* in the *Kōya* manuscript is translated as zero-marking in the *Amakusa* edition. This means that, compared with the *Kōya* manuscript, the *Amakusa* edition tends to make explicit that a given noun phrase is the topic of the sentence."

At approximately the same historical stage, *=de aru* underwent phonological reduction. Frellesvig (2010: 341–342) states that the late LMJ copular paradigm included "new secondary forms which arose in late LMJ: *dyar-* (< *de ar-*)" and that "The nonpast forms in the LMJ paradigm are abbreviated reflexes of adnominal secondary forms, with loss of final *-ru*", specifically "*dya* < *dyaru* < *de aru*" and "*dea* < *de aru*". More generally, in §12.4.1, *Loss of -ru*, Frellesvig (2010: 353) notes that "During LMJ a number of grammatical forms built on or diachronically incorporating *ar-* dropped the final *-ru* in reflexes of the adnominal form", listing "*dya* < *dyaru* < *de aru*" as one such outcome. Finally, he represents the contraction giving rise to the eastern-dialect "forms in *da(-)*" explicitly as "*de aru* > *dea* > *da*", in contrast with the western *de aru* > *dyaru* > *dya* pattern (Frellesvig 2010: 394–395, 400).

Thus, **the increasing formal explicitness of topic marking with =*wa*** documented by Kitazaki (2024) for the *Amakusa* of edition *Heike monogatari* and **the loss of final *-ru* in *de aru*** stated by Frellesvig (2010) **are both situated in late LMJ**. This chronological correspondence does not by itself establish a causal relation, but it is consistent with the possibility suggested above that the discourse-presupposition of the existence of *A*'s referent, increasingly made explicit by *A=wa*, contributed to the informational redundancy and subsequent reduction of sentence-final *aru*. This possibility may be investigated further by cross-tabulating the marking of *A* — *=wa*, *=ga* or *=Ø* — with the predicate forms *=de aru*, *=dyaru*, *=dya*, *=dea* or *=da* within the same texts, while controlling for relevant discourse-pragmatic and textual variables.

Returning to the reduction of the underlying *=ni-te ar-0* to the surface *=da*, via *=de ar-0*, it can be accounted for by the fixed order of these concatenated morphemes and its frequency. In existence-introducing and existence-domain-specifying constructions, the morphemes *=ni*, *-te* and *ar-* — which as a whole corresponds to the schematic expression E( ) ∧ IN(evE( ),) and in which *=ni*, *-te* and *ar-* encode IN(ev( ),), ∧ and E( ), respectively — is invariably ordered in this way, except in scrambling constructions — or *kakimazekōbun* in Japanese linguistics — such as *B**=de** watasi=wa / watasitati=wa / wareware=wa **ar**i-tai* '(lit.) as B, I / we / we want to exist'. **The very high frequency of this fixed order in actual utterances is assumed to have led to the phonological reduction of *=ni-te ar-i* → *=de ar-u* → *=de-a* → *=d-a***, just as in the German and Portuguese examples mentioned above, and likewise in the contractions *si-te=i-ru* → *si-te-ru* 'be doing something' or 'have done sth.', and *si-te=sima-u* → *si-t-sima-u* → *si-t-sya-u* 'end up doing sth.' or 'do sth. inadvertently or unintentionally', as reflected in the synchronically available colloquial forms of Contemporary Japanese.

It is not uncommon for a phonological form whose linear segmentation is no longer transparent to encode multiple meanings. In French, for instance, the present indicative first-person singular form *ai* /e/ of the verb *avoir* 'to have', as in *j'ai* 'I have' — derived from the same inflected form *habeō* /hab-ē-ō/, plausibly surfacing as [ˈha.be.oː], from the Classical Latin *habēre* '*idem.*' — expresses the lexical meaning 'have' in addition to a bundle of functional meanings such as [−non-assertive], [−past], [−posterior], [−anterior], [−non-participant], [−addressee] and [−plural]. Japanese *=da*, too, is nothing more than the outcome of diachronic phonological attrition: four independent morphemes *=ni*, *-te*, *ar-* and *-ru* have been worn down over time and have come to assume a simple shape *=da*.

In sum, ***=da*** as a whole should not be regarded as a copula in the strict sense, but rather as a **phonologically reduced surface realisation of the underlying string *=ni-te ar-0***. In this underlying structure, *=de* — underlyingly *=ni-te* — constitutes an adjunctive postposition–

conjunction complex that conjunctively specifies the conceptual domain in which the theme's existence is located, while *ar-* functions as the one-argument existential predicate and *-0* realises the phonologically null [−past] tense suffix T⁰Ø. This is precisely why the following highly colloquial dialogue in (73) is possible:

(73) *Kare=tte gakusei=**da** yone? — Iya, gakusei=tte yū=ka, gakusei=**de**=mo **aru**=tte kanzi kana.*

'He's a student, right?' — 'Well, if you can call him a student, I'd say he's *also* a student, kind of.'

In (73), the interlocutor, recognising that the underlying structure of *gakusei=da* is *gakusei=de aru*, when asked to confirm that *kare*'s conceptual domain of existence can be identified with that of 'student', replies that it extends to other domains as well, expressed by the form *gakusei=de=**mo** aru*, representable as E(he) ∧ IN(evE(he), student) ∧ IN(evE(he), something else)….

11. Syntactic and logical-propositional differences between NPs with =*da*, nominal adjectives in -*da* and verbal adjectives

Japanese **nominal adjectives** (**NAs**), also referred to as ***na*-adjectives** due to their attributive forms ending in *-na* — the so-called *rentaikei* in Japanese linguistics — as in *yukai-**na** hito* 'a pleasant person', have often been analysed as consisting of "so-called NA stems plus the so-called copula *=da*", in parallel with "NPs plus the so-called copula *=da*", as in *sore=wa* ***yukai=da*** 'it is pleasant', in parallel with *sore=wa* ***gakkō=da*** 'it is a school'.

According to Makino (2025²: 22), however, **the core of NAs lies in the NA stem in *-ni***, as in *yukai-ni* 'be pleasant' and *suki-ni* 'like'. The element *-ni* is not a postposition but the NA thematic — or, stem-forming — suffix, which functions as an indispensable morphological and semantic constituent in the formation of the NA stem. These NA stems are lexical predicates that select arguments — a theme in *yukai-ni*, and an experiencer–theme pair in *suki-ni* — just as verbal-adjective (VA) stems are, including the one-argument predicate *tanosi-ku* 'be enjoyable' with a theme and the two-argument predicate *hosi-ku* 'want' with an experiencer–theme pair. However, outside small clauses functioning as complements of cognition or perception verbs, as in *watasi=wa* [*sore=o yukai-**ni***] *omot-ta / kanzi-ta* 'I found / felt [it pleasant]', such NA stems cannot function as predicates in their bare *-ni* form unlike VA stems, as in *sore=wa tanosi-**ku**, *yukai-**ni*** 'it is enjoyable, pleasant' and *sore=wa *yukai-**ni**, tanosi-**i*** 'it is pleasant, enjoyable'.

Consequently, an NA stem forms a nominal-adjective phrase containing PRO, as in [*PRO yukai-ni*], which serves as the complement of the adjunctive conjunction *-te*, yielding a ConjP [*PRO yukai-ni-te*]. This ConjP is then adjoined to the VP headed by the existential-verb stem *ar-* within a one-argument obligatory construction, as in [*sore*$_i$*=wa t*$_i$ *ar-0-0*] 'it exists', where *-0-0* represents the sequence of T^{0}Ø and Top^{0}Ø. Only in this configuration can the NA stem function as the predicate of an independent sentence, as in [*sore*$_i$*=wa* [*PRO*$_i$ *yukai-ni-te*] *t*$_i$ *ar-0-0*], which surfaces as *sore=wa yukai-d-a* [*sore*$_i$*=wa* [*PRO*$_i$ *yukai-d*] *t*$_i$ *a-0-0*] ''(lit.) it exists, being pleasant'. On the other hand, the so-called copular construction *sore=wa gakkō=d-a* '(lit.) it exists, with its existence located in the conceptual domain of "school"' can be underlyingly analysed, as already discussed, as [*sore*$_i$*=wa* [*gakkō=ni-te*] *t*$_i$ *ar-0-0*], which surfaces [*sore*$_i$*=wa* [*gakkō=d*] *t*$_i$ *a-0-0*].

Thus, the so-called copular sentences and the nominal-adjectival sentences share the syntactic property that they both consist of an existence-introducing, one-argument core construction combined with either an existence-domain-specifying adjunctive ConjP in the former or a state-specifying adjunctive ConjP in the latter, and can both be schematically represented as conjunctive propositions, as in (74) and (75):

(74) *Sore=wa gakkō=d-a.* [*sore*$_i$*=wa* [*gakkō=ni-te*] *t*$_i$ *ar-0-0*] E(sore) ∧ IN($_{\text{ev}}$E(sore), gakkō)

(75) *Sore=wa yukai-d-a.* [*sore*$_i$*=wa* [*PRO*$_i$ *yukai-ni-te*] *t*$_i$ *ar-0-0*] E(sore) ∧ PLEASANT(sore)

The difference is that, since the PP [*gakkō=ni*] in (74) does not constitute a lexical predicate, it does not select any argument; therefore, the adjunctive construction [*gakkō=ni-te*] does not contain PRO, unlike the adjunct [*PRO yukai-ni-te*] in the nominal-adjectival construction in (75).

By contrast, **verbal adjectives (VAs)** — also referred to as adjectives, canonical adjectives or ***i*-adjectives** due to their attributive forms ending in *-i*, as in *tanosi-**i** hito* 'a fun person' — form constructions in which they themselves function as the main predicate. Such constructions, as in *sore=wa tanosi-i* 'it is enjoyable', consist only of a state-specifying obligatory argumental construction without any adjunct, and thus can be represented as atomic propositions, as in (76):

(76) *Sore=wa tanosi-i.* [*sore*$_i$*=wa t*$_i$ *tanosi-i-0-0*] ENJOYABLE(sore)

The syntactic and logical-propositional differences between (74) and (75), on the one hand, and (76), on the other, become manifest in their default negation patterns, as shown in (77) – (79):

(77) *Sore=wa gakkō=de=**wa*** ***na-i***. E(sore) ∧ ¬**IN(evE(sore), gakkō)**

(78) *Sore=wa yukai-de=**wa*** ***na-i***. E(sore) ∧ ¬**PLEASANT(sore)**

(79) *Sore=wa tanosi-ku* ***na-i***. **¬ENJOYABLE(sore)**

The default negation patterns of the constructions in (74) and (75) — representable as the conjunctive propositions E(sore) ∧ IN(evE(sore), gakkō) and E(sore) ∧ PLEASANT(sore) — are partial, namely E(sore) ∧ ¬IN(evE(sore), gakkō) and E(sore) ∧ ¬PLEASANT(sore), as in (77) and (78), whereas that of the construction in (76) — representable as the atomic proposition ENJOYABLE(sore) — is total, namely ¬ENJOYABLE(sore), as in (79).

It follows that the stem allomorph *at-* of the verb *ar-u* occurring in the past form *tanosi-k-**at**-ta*, for example, does not instantiate the one-argument existential verb *ar-* 'exist' used in both so-called copular and nominal-adjectival sentences. Rather, as stated in Makino (2025[1]: 4), it is an allomorph of "the stem *ar-* from the light verb *ar-u*" inserted as a matter of formal necessity in order to concatenate the **verbal** past-tense suffix *-ta* with the **non-verbal** VA stem *tanosi-ku*, which is incompatible with the former, as in *tanosi-ku* + *-ta* → *tanosi-ku* + ***ar-*** + *-ta* → *tanosi-k-**at**-ta*. The default negation of this past form *tanosi-k-at-ta* is likewise total, namely *tanosi-ku **na-k**-a-ta*, rather than the partial negation *tanosi-ku**=wa*** ***na-k**-at-ta*. This demonstrates that *tanosi-ku* does not constitute an adjunct forming a separate proposition and thus provides further evidence that *ar-* in this construction is not a one-argument existential verb. Note finally that the partial negation of the construction in (80) gives rise to an implication such as ∧ SOME-OTHER-STATE(sore):

(80) *Sore=wa tanosi-ku=**wa*** ***na-i***. ¬ENJOYABLE(sore) ∧ SOME-OTHER-STATE(sore)

The syntactic derivation of the pseudo-copular, nominal-adjectival and verbal-adjectival constructions in (81) – (83) (= (74) – (76)) and (84) can be illustrated as follows:

(81) *Sore=wa gakkō=d-a.* '(lit.) It exists, with its existence located in the domain of "school".'

Adjunct	**Core construction**
[***=ni***]	[***ar-***]
→ [***gakkō***=*ni*]	→ [***sore*** *ar-*]
→ [[*gakkō=ni*] ***-te***]	
	→ [**[[*gakkō=ni*] *-te*]** [*sore ar-*]] (adjunct embedding)
	→ [[[[*gakkō=ni*] *-te*] [*sore ar-*]] ***T^0Ø***]
	→ [[[[[*gakkō=ni*] *-te*] [*sore ar-*]] *T^0Ø*] ***Top^0Ø***]
	→ [***sore***$_i$***=wa*** [[[[[*gakkō=ni*] *-te*] [***t***$_i$ *ar-*]] *T^0Ø*] *Top^0Ø*]]
	→ *sore=wa gakkō=**d**-a*

As already noted in the previous section, the underlying sequences *=ni-te-ar-0* and *-ni-te-ar-0* surfaces as *=d-a* and *-d-a* in (81), as well as in (82) below:

(82) *Sore=wa yukai-d-a.* '(lit.) It exists, being pleasant.'

Adjunct	**Core construction**
[***yukai-ni***]	[***ar-***]
→ [***PRO*** *yukai-ni*]	→ [***sore*** *ar-*]
→ [[*PRO yukai-ni*] ***-te***]	
	→ [**[[*PRO*$_i$ *yukai-ni*] *-te*]** [*sore*$_i$ *ar-*]] (adjunct embedding)
	→ [[[[*PRO*$_i$ *yukai-ni*] *-te*] [*sore*$_i$ *ar-*]] ***T^0Ø***]
	→ [[[[[*PRO*$_i$ *yukai-ni*] -te] [*sore*$_i$ *ar-*]] *T^0Ø*] ***Top^0Ø***]
	→ [***sore***$_i$***=wa*** [[[[[*PRO*$_i$ *yukai-ni*] *-te*] [***t***$_i$ *ar-*]] *T^0Ø*] *Top^0Ø*]]
	→ *sore=wa yukai-**d**-a*

(83) *Sore=wa tanosi-i.* 'It is enjoyable.'

Core construction

[***tanosi-ku***]

→ [***sore*** *tanosi-ku*]

→ [[*sore tanosi-ku*] ***T*^{0}*Ø***]

→ [[[*sore tanosi-ku*] *T*^{0}*Ø*] ***Top*^{0}*Ø***]

→ [***sore*$_{i}$*=wa*** [[[***t*$_{i}$** *tanosi-ku*] *T*^{0}*Ø*] *Top*^{0}*Ø*]]

→ *sore=wa tanosi-**i*** (allomorphy or amalgam)

In the VA non-past form in (83), the sequence [*-ku* + *T*^{0}*Ø*] is morphophonologically realised as *-i*, yielding the surface form *tanosi-i* from the underlying elsewhere form *tanosi-ku*. The exponent *-i* may therefore be analysed either as a contextual allomorph *-i* of the VA thematic suffix *-ku* in the non-past environment or, alternatively, as an amalgamated realisation of the sequence [*-ku* + *T*^{0}*Ø*], as stated in Makino (2025[1]: 14).

(84) *Sore=wa tanosi-k-at-ta.* 'It was enjoyable.'

Core construction

[***tanosi-ku***]

→ [***sore*** *tanosi-ku*]

→ [[*sore tanosi-ku*] ***-ta***]

→ [[*sore tanosi-ku*] ***ar-*** *-ta*] (light-verb-stem insertion)

→ [[[*sore tanosi-ku*] *ar- -ta*] ***Top*^{0}*Ø***]

→ [***sore*$_{i}$*=wa*** [[[***t*$_{i}$** *tanosi-ku*] *ar- -ta*] *Top*^{0}*Ø*]]

→ *sore=wa tanosi-**k**-a**t**-ta*

In the analysis of the VA past form in (84), the choice between the non-past T^{0}Ø, realised as *-0*, and the past suffix *-ta* is treated as a paradigmatic selection made at a single derivational stage, since the [−past] *-0* and the [+past] *-ta* constitute a paradigm whose terms acquire linguistic value solely through their opposition at the same point of speech. When the past suffix *-ta* is selected, it proves incompatible with non-verbal predicates such as *tanosi-ku*, thereby triggering the insertion of the light verb stem *ar-* as a repair strategy. Then, the stem-final vowel *-u* of the suffix *-ku* is deleted before *ar-*, together with the assimilation of *ar-* to *at-* before *-ta*.

Finally, the more abstract pseudo-copular, nominal-adjectival and verbal-adjectival constructions corresponding to (81) – (83) — namely, ***A=wa B=d-a***, ***A=wa B-d-a*** and ***A=wa B-i***, where *A* designates an NP and *B* designates an NP, an NA radical and VA radical, respectively — can be schematically represented as in (85) – (87):

(85) *A=wa B=d-a*:	E(A) ∧ IN(evE(A), B)	(e.g. *sore=wa gakkō=d-a*)
(86) *A=wa B-d-a*:	E(A) ∧ B-ni(A)	(e.g. *sore=wa yukai-d-a*)
(87) *A=wa B-i*:	B-ku(A)	(e.g. *sore=wa tanosi-i*)

12. Classical Latin *esse*, Japanese *ari/aru* and Old English *Wesan/Bēon* — an exploratory note

Japanese is not the only language that employs a one-argument existential verb in a copula-like way. **In Classical Latin, like Japanese, the default — namely, the most unmarked — one-argument existential verb was employed to encode a relation between an entity and its specification as to what it is**. In Latin, the verb *esse* 'to exist; to be' was used both as a one-argument existential verb and as a so-called copula, as exemplified in *dī sunt* 'the gods exist'[6] with a topicalised definite subject *dī* and with no locative phrase, and *dī testēs sunt* 'the gods are witnesses'.

In the latter construction, the NP *testēs* 'witnesses' is traditionally analysed as a *copular*

[6] Cicero, *De divinatione* 2.49 contains a clear example of the one-argument existential *esse* 'exist' with no locative phrase and with a topicalised subject NP *dī* at a clause-initial position: ***nōn** igitur **dī sunt*** 'therefore **the gods do not exist**'. In a Stoic argument, the NP *dī* is first introduced as a discourse referent in the protasis *sī sunt dī* ... 'if there are gods …' or 'if gods exist …'. It is then maintained as a null subject *pro* along a chain of coordinated clauses. The conclusion *nōn igitur dī sunt* 'therefore the gods do not exist' then restates *dī* overtly at the point of inference, where it is naturally interpreted as referential given information. Thus, the constructed example *dī sunt* 'the gods exist' here is likely to be acceptable in this sense. On the other hand, the clause *dī testēs sunt* is borrowed from T. Livius's *Ab urbe condita* 35.19: *… pater Hamilcar et **dī testēs sunt*** '… my father Hamilcar and the gods are witnesses'.

predicate nominative, whereas here it is proposed that this NP may instead be analysed as an ***appositive* predicate nominative to the subject NP** *dī* 'the gods', **marked by the same nominative case**, likewise as in the following example: *dēfendī rem pūblicam* ***adulēscēns****, nōn dēseram* ***senex*** ' I, as a young man, defended the republic, I, as an old man, shall not desert it' (Cic. *Phil.* 2.118). In its primitive sense, the construction *dī testēs sunt* can thus be interpreted as 'the gods, being witnesses, exist' or 'the gods, as witnesses, exist', representable as EXIST(dī) ∧ PRED(dī, testēs), where PRED abbreviates 'predication' and will be more simply abbreviated as P below. If the initial conjunct E(dī) is deleted, the remaining proposition P(dī, testēs) will constitute what is traditionally described as a copula. (It should be noted that the term "appositive predicate nominative" is used here for the nominative nominal *testēs* in *dī testēs sunt*, which is analysed as predicative apposition rather than as a copular complement.)

In Classical Latin, however, the E(A) component in the conjunctive proposition **E(A)** ∧ P(A, B) appears to have remained present, albeit typically backgrounded, as in the Japanese structure **E(A)** ∧ IN(evE(A), B) discussed in Sections 5 and 10, since, to express 'to want to be/become', the state-change verb *fierī* 'to become' tended to be used rather than *esse*, as in ***augur*** *posteā* ***fierī voluī****, quod anteā neglēxeram* '(lit.) later I wanted to become an augur, having previously neglected it' (Cic. *Fam.* 15.4), as opposed to ***cupiō****, patrēs cōnscrīptī,* ***mē esse clēmentem*** '(lit.) I desire, senators, to exist as merciful' (Cic. *Catil.* 1.4). This opposition can be schematically represented as in:

(88) *A B* ***fierī*** *cupit/vult.* 'A wants to **be** B.': A WANT [BECOME(A, B)]

(89) *A B* ***esse*** *cupit/vult.* 'A wants to **exist as** B.': A WANT [E(A) ∧ P(A, B)]

The relation between (88) and (89) is closely paralleled by the following opposition between (90) and (91) in Japanese, discussed in Section 6:

(90) *A=wa B=ni* ***nari****-tai.* 'A wants to **be** B.': A WANT [BECOME(A, B)]

(91) *A=wa B=de* ***ari****-tai.* 'A wants to **exist as** B.': A WANT [E(A) ∧ IN(evE(A), B)]

In Classical Latin, the tendency to use *fierī* 'to become' rather than *esse* to express 'to want to be' is compatible with the view that, even in its copula-like use, the component E(A) still remained present. Thus, Japanese and Classical Latin share the commonalities in (92) and (93):

(92) They employ existing linguistic resources in encoding a relation between an entity and its specification as to what it is.

(93) One of those resources is the default one-argument existential verb — namely, *ar-u* in Japanese and *esse* in Classical Latin.

On the other hand, they differ in (94) – (96):

(94) The other resources differ — namely, the conceptual-locative postposition *=ni* and the adjunctive-conjunction suffix *-te* in Japanese, and predicative apposition and agreement in Classical Latin.

(95) In Classical Latin, the component E(A) could, in principle, be deleted from E(A) ∧ P(A, B) without syntactic and logical difficulty in predicating B of A, thereby yielding a genuine copula P(A, B), as in E(A) ∧ P(A, B) → P(A, B) — even though, in practice, E(A) does not appear to have been deleted.

(96) In Japanese, by contrast, the component E(A) cannot, as a matter of syntactic and logical structure, be deleted from E(A) ∧ IN(evE(A), B), since, if E(A) were deleted, *A=wa B=de aru* would be representable as IN(ev( ), B) through the deletion of its existential meaning from *ar-*, which would be logically incomplete, and syntactically inadequate since *A* would not be assigned any θ-role by the semantically bleached *ar-*.

Later, the deletion of the E(A) component from *esse* — that is, E(A) ∧ P(A, B) → P(A, B) — may be analysed as having given rise to a genuinely copular use P(A, B). In parallel its existential use E(A) may have receded, with other expressions of E(A) subsequently developing in the Romance languages. Finally, it is suggestive that **the English copula *be* — in Old English, *wesan/bēon* — was used, like Latin *esse* and Japanese *ari/aru*, as the default one-argument existential verb**, as in *þa hwile þe he* ***wæs*** 'while he existed' (Thorpe 1865: 167, l.9) [= *ðā hwīle ðe hē wæs*]; *he bið á* ***wesende*** 'he is ever existent' (Morris 1880: 19, ll.25–26) [= *hē bið ā wesende*][7].

[7] See also Stevick (1992: 149). In Hall (1916), the verbs that include a sense glossed as "*exist*" are *bēon* (2), *faran* (8), *libban* (3), *standan* (11), *wunian* (4), where the numerals in parentheses indicate the sense number under which "exist" is listed. The other verb corresponding — alongside *bēon* — to Modern English *be*, namely *wesan*, is glossed only as "*to be*" and "*happen*" in Hall, though the entry contains a

13. Conclusion

In summary, **Japanese has no genuine copula — that is, *=da* is not *be***. Instead, Japanese is **a language that has employed existence, conjunction and location to encode, compositionally, a relation between an entity and its specification as to what it is**. The function of *=da* and its non-contracted form *=de aru* is to indicate that something **exists** and that this existence is **conjunctively** specified as **located in a conceptual domain** defined by a category. Japanese linguistically encodes E(A) ∧ IN(evE(A), B) through *=ni*, *-te* and *ar-*, whereas any interpretation in terms of A = B or A ∈ B arises as a derived meta-level inference. Thus, treating *=da* as *be* may be practical for pedagogical, translational and surface-level purposes, but **the Japanese grammatical system contains no primitive corresponding to *be***.
Syntactically and semantically, the construction *A=wa B=de aru* consists of an existence-introducing, one-argument obligatory construction [*A=wa aru*] 'A exists' combined with an existence-domain-specifying adjunctive construction [*B=de*] — underlyingly [*B=ni-te*] — meaning 'with (A's existence) located in the conceptual domain of B' or, more practically, 'and it is as B'. In this construction, the most crucial logical — albeit not syntactic — predicate is not the verbal stem *ar-* that introduces *A*'s existence, but rather the predicate-like conceptual-locative postposition *=ni* within the postposition–conjunction complex *=de*, which locates *A*'s existence in the conceptual domain of B.

Consequently, treating *=da*, or its non-contracted form *=de aru*, as a single copula is structurally problematic. The form *=de aru* is more appropriately analysed as an **existence-and-domain complex**, consisting of the one-argument existential verb *aru* 'exist' and the predicate-like conceptual-locative postposition *=ni* 'in' plus the adjunctive-conjunction suffix *-te* 'and', which together surface as *=de* and conjunctively specify the conceptual domain in which the existence of the theme is located. The more colloquial form *=da* is a phonologically contracted

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cross-reference "v. also bēon". When these verbs are examined in Bosworth & Toller (1898), *bēon* is defined (*ibid.*: 85) as "BEÓN [bión], …*To* BE, *exist*, *become*; esse, fieri", with "*exist*" appearing as the second sense, whereas *faran* 'go', *libban* 'live', *standan* 'stand' have no 'exist' sense at all, their basic meanings being those given in single quotation marks here. *Wunian* has the basic meanings "*dwell*" and "*remain*" and only under section III does it include "*exist*", which indicates that 'exist' is not its primary sense. Finally, for *wesan*, which together with *bēon* constitutes the Modern English verb *be*, Bosworth and Toller provide the crucial description (*ibid.*: 1210): "wesan; p. *wæs*, pl. *wǽron To be:*— Wesan and beón *fore*, Wrt. Voc. ii. 34, 61. I. as an independent verb, (1) denoting existence *to be*, *exist*:— Wesendum, beóndum *existentibus*, Wrt. Voc. ii. 32, 63. (a) of animate objects, *to exist, live*". This interpretation is consistent with later scholarship: Mateo Mendaza, R. (2021: 95): "beon and wesan are purely existential verbs"; Cichosz et al. (2022: 87): No.1 *wesan* (translation *to be*); No.2 *beon* (translation *to be*) in "Table 12. 15 most frequent Old English non-lexical verbs". On the basis of the evidence reviewed above, it may be concluded that the default existential verb in Old English was *wesan/bēon*.

— though pragmatically and informationally motivated — form corresponding to the more formal and literal *=de aru*.

Thus, through the above morphosyntactic and semantic analysis, it has been possible to account, by means of a single, consistent principle — namely, the **introduction of existence by [NP_1 *ar-*]** and the **conjunctive specification of the conceptual domain of existence by [NP_2=*ni-te*] surfacing as [NP_2=*de*]**, schematically representable as the conjunctive proposition **$E(NP_1) \wedge IN(evE(NP_1), NP_2)$, not reducible to the atomic $BE(NP_1, NP_2)$** — for:

- Morphosyntactic separability of *de* and *aru* in *=de aru*
- Syntactic and prosodic constituency of *=de* and the NPs preceding it
- Prosodic-word boundary between *=de* and *aru*
- Contextually optional omission of *aru* in *=de aru*
- Evidence for analysing *=de* as an adjunct-marking postposition–conjunction complex *=ni-te* that functions as an eventuality-domain specifier $\wedge$ (ev( ), NP)
- Foregrounding of the lexical meaning 'exist' of the verb *aru*
- Grounds on which the Japanese desiderative *=de ari-tai* is not equivalent to the English *to want to be*
- Diachronic split from Old to Modern Japanese between *A B=ni ari → A B=nite ari → A=wa B=de aru* and *A B=ni ari → A=wa B=ni aru/iru*
- Evidence that the default negation of *A=wa B=de aru* is not the total *A=wa B=de nai* but the partial *A=wa B=de=wa nai*
- Syntactic and information-structural differences between *A=wa B=de aru* and *A=ga B=de aru*
- Underlying structure *=ni-te ar-T⁰Ø* surfacing as *=d-a-T⁰Ø*
- Pragmatic and informational motivation for the phonological reduction *aru → a* in *=da*
- Commonalities and differences between the pseudo-copular, nominal-adjectival and verbal-adjectival constructions

Put differently, to maintain the claim that *=da* is *be*, one would have to assume that the inherent meanings of *=ni*, *-te* and *ar-* — which together still autonomously underlie *=da* — have been eliminated or, alternatively, somehow changed into another meaning. One would then further need to explain when these meanings were eliminated or changed, why they were eliminated or changed, at what grammatical level they were eliminated or changed, and why meanings that are assumed to have been eliminated or changed nevertheless become foregrounded in desiderative constructions, exert structural effects under negation, create prosodic boundaries and maintain diachronic continuity.

Finally, while the present study reinterprets *=da* and *=de aru* as a postposition–conjunction–verb complex encoding $E(A) \wedge IN(evE(A), B)$, rather than as a copular predicate representable as $BE(A, B)$, this proposal is not intended to invalidate previous research conducted within the copular framework. Much of the empirical and descriptive work on so-called copular sentences remains valuable and can be brought into dialogue with the present account as work bearing on the $IN(evE(A), B)$ component, that is, as research into how the conceptual domain of existence is structured. In this respect, the present analysis may be regarded not as a rejection of the copular tradition but as an attempt to reconsider some of its insights within a different model of Japanese predicational structure. Furthermore, the interaction between the existence-introducing component $E(A)$ and the domain-specifying component $IN(evE(A), B)$, which has not always been foregrounded in previous research, may provide a useful perspective for future studies on how these two semantic components relate to one another in information structure, negation and discourse.

References

Bosworth, J., & Toller, T. N. (1898). *An Anglo-Saxon dictionary*. Oxford: Clarendon Press.

Cicero, M. T. (1917). *De natura deorum* (O. Plasberg, Ed.). Lipsiae: B. G. Teubner.

Cicero, M. T. (1988). *Epistulae ad familiares: Libri I–XVI* (D. R. Shackleton Bailey, Ed.). Stutgardiae: B. G. Teubner.

Cicero, M. T. (2009). *In Catilinam I–IV* (J. F. Stout, Ed.). Bristol, England: Bristol Classical Press.

Cicero, M. T. (1986). *In M. Antonium orationes Philippicae XIV* (P. Fedeli, Ed., 2nd ed.). Lipsiae: B. G. Teubner.

Cichosz, A., Pęzik, P., Grabski, M., Karasińska, S., Adamczyk, M., Rybińska, P., & Ostrowska, A. (2022). *A frequency dictionary of Old English prose for learners of Old English and historical linguists*. Łódź University Press. https://doi.org/10.18778/8220-899-3.

Dalla Chiesa, S. (2015). Japanese case marker de in copular sentences: Essive or locative? *Journal of Foreign Language Teaching and Applied Linguistics, 1*(3), 1–11. https://doi.org/10.14706/JFLTAL15131.

Frellesvig, B. (2010). *A History of the Japanese Language*. Cambridge University Press.

Hall, J. R. C. (1916). *A concise Anglo-Saxon dictionary: For the use of students* (2nd ed., revised and enlarged). London: The Macmillan Company.

Hasegawa, N. (1999). *Seisei Nihongogaku Nyūmon* [An introduction to generative Japanese linguistics]. Tokyo: Taishūkan Shoten.

Kitazaki, Y. (2024). *Heike monogatari* no taishō kōpasu o tōshite miru chūsei nihongo [Late Middle Japanese through the parallel corpus of *The Tales of the Heike*]. *Mathematical Linguistics, 34*(7), 482–497. https://doi.org/10.24701/mathling.34.7_482.

Livius, T. (1991). *Ab urbe condita: Libri XXXI–XL* (J. Briscoe, Ed.). Stuttgart: B. G. Teubner.

Makino, S. (2025[1]). *A New Analysis of the Morphological Structure of Japanese Verbal Adjectives — KU-adjectives.* Preprint. https://doi.org/10.5281/zenodo.15764311.

Makino, S. (2025[2]). *An Alternative Analysis of the Morphosyntactic and Semantic Structure of Japanese Nominal Adjectives — NI-adjectives.* Preprint. https://doi.org/10.5281/zenodo.17254868.

Mateo Mendaza, R. (2021). Identifying the Old English exponent for the semantic prime LIVE. *SELIM. Journal of the Spanish Society for Medieval English Language and Literature, 26*(1), 85–107. https://doi.org/10.17811/selim.26.2021.85-107.

Matsuoka, M. (2019). On the locative structure of -te iru progressives in Japanese. *Gengo Kenkyu, 155*, 131–157. https://doi.org/10.11435/gengo.155.0_131.

Miyagawa, S., & Saito, M. (Eds.). (2008). *The Oxford Handbook of Japanese Linguistics*. Oxford: Oxford University Press.

Miyaoka, O. (2002). *Go to wa Nani ka: Esukimōgo kara Nihongo o miru* [What is a Word: viewing Japanese from the perspective of Eskimo languages]. Tokyo: Sanseido.

Morris, R. (Ed.). (1880). *The Blickling homilies of the tenth century: From the Marquis of Lothian's unique MS. A.D. 971*. London, England: Printed for the Early English Text Society by N. Trübner.

Nakayama, T. (1987). *Hyōzyun Raten Bunpō* [Classica Grammatica Latina. Tokyo: Hakusuisha.

Nakayama, T. (2007). *Koten Ratengo Bunten* [Ars Latina]. Tokyo: Hakusuisha.

Nespor, M., & Vogel, I. (2007). *Prosodic phonology: With a new foreword*. Berlin: Mouton de Gruyter.

Noya, S. (1994). *Ronrigaku* [Logic: An introduction]. Tokyo: University of Tokyo Press.

Oda, M. (2003). *Kodai Nihongo Bunpou* [Ancient Japanese Grammar]. Tokyo: Chikuma Shobo.

Okimori, T. (2017). *Nihongo Zenshi* [A Comprehensive Chronological History of the Japanese Language]. Tokyo: Chikuma Shobo.

Ōno, S., Satake, A., & Maeda, K. (Eds.). (1990). *Iwanami Kogo Jiten* [Iwanami Dictionary of Old Japanese] (revised ed.). Tokyo: Iwanami Shoten.

Stevick, R. D. (1992). *A First Book of Old English*. Tokyo: Kenkyusha.

Thorpe, B. (Ed.). (1865). *Diplomatarium Anglicum Ævi Saxonici*: A collection of English charters from the reign of King Æthelberht of Kent to that of William the Conqueror. London: Macmillan.

Tsuji, Y. (Ed.). (2002). *Ninchi Gengogaku Kīwādo Jiten* [Dictionary of Key Terms in Cognitive Linguistics]. Tokyo: Kenkyusha.

Tsujimura, N. (Ed.). (2002). *The Handbook of Japanese Linguistics*. Malden, MA: Blackwell Publishing.

Tsujimura, N. (2007). *An Introduction to Japanese Linguistics* (2nd ed.). Malden, MA: Blackwell Publishing.